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Energy-supply system for supplying energy to an electrical load from a battery

Abstract

A motorized window treatment may be configured to adjust a position of a covering material to control the amount of daylight entering a space. The motorized window treatment may include a DC power source for charging an energy storage element, such as a supercapacitor and/or rechargeable battery. The energy storage element may be configured to provide power for the operation of a motor used to adjust the position of the covering material. The energy storage element may discharge when providing power to the motor and may charge such that the current it draws from a battery is at a desired average current level that extends the lifetime of the battery.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of Provisional U.S. Patent Application No. 63/265,671, filed Dec. 17, 2021, and Provisional U.S. Patent Application No. 63/266,824, filed Jan. 14, 2022, the entire disclosures of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

(1) A typical window treatment, such as a roller shade, a drapery, a roman shade, and/or a venetian blind, may be mounted in front of a window or opening to control an amount of light that may enter a user environment and/or to provide privacy. A covering material (e.g., a shade fabric) on the window treatment may be adjusted to control the amount of daylight from entering the user environment and/or to provide privacy. The covering material may be manually controlled and/or automatically controlled using a motorized drive system to provide energy savings and/or increased comfort for occupants. For example, the covering material may be raised to allow light to enter the user environment and allow for reduced use of lighting systems. The covering material may also be lowered to reduce the occurrence of sun glare.

SUMMARY

(2) A motor drive unit for a motorized window treatment may include a bus capacitor configured to store a bus voltage, and a motor configured to control movement of a covering material of the motorized window treatment. The motor drive unit may also include a motor drive circuit configured to receive the bus voltage and conduct a motor current through the motor for controlling power delivered to the motor to control movement of the covering material. The motor drive unit may include a first power source configured to generate a first power source voltage, and a second power source configured to generate a second power source voltage. In some examples, the motor drive unit does not include the first power source, since for example, the first power source may be purchased and installed by the user after purchase. The first power source may comprise batteries, and the second power source may comprise an energy storage element. The batteries may be comprised of a first battery chemistry, and the energy storage element may be comprised of a second battery chemistry, where the first battery chemistry is different from the second battery chemistry. For example, the one or more batteries comprise alkaline batteries, such as off-the-shelf alkaline batteries that are replaceable by the user, and in some examples, rechargeable. The energy storage element comprises one or more lithium batteries and/or one or more supercapacitors.
(3) The motor drive unit may include a first switching circuit coupled between the first power source and the bus capacitor. The motor drive unit may also include a second switching circuit coupled between the second power source and the bus capacitor. The first switching circuit comprises a field-effect transistor (FET), or wherein the second switching circuit comprises a FET.
(4) The motor drive unit may also include a control circuit configured to, prior to controlling the motor drive circuit to generate the motor voltage across the motor to control the movement of the covering material, gradually close the first switching circuit to charge the magnitude of the bus voltage to the magnitude of the first power source voltage when the magnitude of the second power source voltage is less than the threshold, and control the motor drive circuit to conduct the motor current from the first power source and through the motor to control the movement of the covering material. The control circuit may be configured to, prior to controlling the motor drive circuit to generate the motor voltage across the motor to control the movement of the covering material, close the second switching circuit to charge the magnitude of the bus voltage to the magnitude of the second power source voltage when a magnitude of the second power source voltage is greater than a threshold, and control the motor drive circuit to conduct the motor current from the second power source and through the motor to control the movement of the covering material. In some examples, the threshold may indicate a storage level sufficient to complete a full movement of the covering material from a fully-lowered position to a fully-raised position. In some examples, the threshold may vary depending on an amount of movement of the covering material required by a

received command.
(5) The control circuit may also be configured to open at least one of the first switching circuit or the second switching circuit that was closed to control the motor when movement of the covering material is complete. For instance, the motor drive unit may be configured such that the first switching circuit and the second switching circuit cannot both be closed at the same time.
(6) To gradually close the first switching circuit, the control circuit may be configured to generate a pulse width modulated (PWM) gate signal at a gate of the first switching circuit. For example, the control circuit may be configured to increase the on-time of the PWM gate signal from one period to the next while gradually closing the first switching circuit. The control circuit may be configured to generate the PWM gate signal to close the first switching circuit using open-loop control. Further, in some examples, to close the second switching circuit, the control circuit may be configured to pulse width modulate a first switch control signal, wherein the first switch control signal is configured to render the first switching circuit conductive and non-conductive.
(7) In some examples, to gradually close the first switching circuit, the control circuit may be configured to decrease an impedance of the first switching circuit from a non-conductive impedance to a conductive impedance. The non-conductive impedance of the first switching circuit may be greater than the conductive impedance of the first switching circuit. To gradually close the first switching circuit, the control circuit may be configured to control an average impedance of the first switching circuit to increase from zero to 100%. In some examples, the first switching circuit may include a field-effect transistor (FET), and, to gradually close the first switching circuit, the control circuit may be configured to control an impedance of the FET of the first switching circuit in a linear region.
(8) The motor drive unit may include a filter circuit, such as an inductor, coupled in series between the first switching circuit and the bus capacitor. The filter circuit may be configured to filter the motor current conducted through the first power source when the first switching circuit is conductive and the motor drive circuit is controlling the power delivered to the motor. The filter circuit may be configured to filter the motor current to conduct a first power source current through the first power source that has a DC magnitude. The motor drive unit may include a diode coupled between circuit common and the junction of the first power source current and the filter circuit. The diode may be configured to conduct current through the inductor and the bus capacitor when the first power source current is non-conductive while the first power source current is gradually closed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1**A and **1**B depict an example motorized window treatment that includes a roller tube and a flexible material windingly attached to the roller tube.
(2) FIG. **2**A is a perspective view of an example battery-powered motorized window treatment with the batteries removed.
(3) FIG. **2**B is another perspective view of the example battery-powered motorized window treatment shown in FIG. **2**A with the batteries removed.
(4) FIG. **3**A is a front cross-section view of another example battery-powered motorized window treatment taken through the center of a roller tube of the motorized window treatment.
(5) FIG. **3**B is a side view of an idler end of the example battery-powered motorized window treatment of FIG. **3**A.
(6) FIG. **4** is a block diagram of an example motor drive unit of a motorized window treatment.
(7) FIG. **5** is a block diagram of an example energy storage element and power converter.
(8) FIG. **6** is an example of waveforms that illustrate an operation of an energy storage element,

power converter, and filter.
(9) FIG. **7** is an example flowchart of a control procedure for charging an energy storage element that may be executed by a control circuit.
(10) FIG. **8** is a flowchart of a control procedure for charging an energy storage element that may be executed by a control circuit.
(11) FIG. **9** is a flowchart of a control procedure for controlling a first switch and/or a second switch for selectively powering an electrical load from one or more batteries or an energy storage device.
(12) FIG. **10** is a flowchart of an example procedure for driving a load by drawing current from one or more batteries or from an energy storage element.
(13) FIG. **11** is a flowchart of an example procedure for driving a load by drawing current from one or more batteries or from an energy storage element.

DETAILED DESCRIPTION

(14) FIGS. **1**A and **1**B depict an example motorized window treatment **100** (e.g., a battery-powered motorized window treatment system) that includes a roller tube **110** and a flexible material **120** (e.g., a covering material) windingly attached to the roller tube **110**. The motorized window treatment **100** may include one or more mounting brackets **130**A, **130**B configured to be coupled to or otherwise mounted to a structure. For example, each of the mounting brackets **130**A, **130**B may be configured to be mounted to (e.g., attached to) a window frame, a wall, or other structure, such that the motorized window treatment **100** is mounted proximate to an opening (e.g., over the opening or in the opening), such as a window for example. The mounting brackets **130**A, **130**B may be configured to be mounted to a vertical structure (e.g., wall-mounted to a wall as shown in FIG. **1**) and/or mounted to a horizontal structure (e.g., ceiling-mounted to a ceiling). For example, the mounting brackets **130**A, **130**B may be rotated 90 degrees from what is shown in FIG. **1**.
(15) The roller tube **110** may operate as a rotational element of the motorized window treatment **100**. The roller tube **110** may be elongate along a longitudinal direction L and rotatably mounted (e.g., rotatably supported) by the mounting brackets **130**. The roller tube **110** may define a longitudinal axis **116**. The longitudinal axis **116** may extend along the longitudinal direction L. The mounting bracket **130**A may extend from the structure in a radial direction R, as shown in FIG. **1**B. It should be appreciated that when the mounting brackets **130** are ceiling-mounted, the mounting bracket **130**A may extend from the structure in a transverse direction T. The radial direction R may be defined as a direction perpendicular to the structure and the longitudinal axis **116**. The flexible material **120** may be windingly attached to the roller tube **110**, such that rotation of the roller tube **110** causes the flexible material **120** to wind around or unwind from the roller tube **110** along a transverse direction T that extends perpendicular to the longitudinal direction L. For example, rotation of the roller tube **110** may cause the flexible material **120** to move between a raised (e.g., open) position (e.g., as shown in FIG. **1**) and a lowered (e.g., closed) position along the transverse direction T.
(16) The roller tube **110** may be made of aluminum. The roller tube **110** may be a low-deflection roller tube and may be made of a material that has high strength and low density, such as carbon fiber. The roller tube **110** may have, for example, a diameter of approximately two inches. For example, the roller tube **110** may exhibit a deflection of less than ¼ of an inch when the flexible material **120** has a length of 12 feet and a width of 12 feet (e.g., and the roller tube **110** has a corresponding width of 12 feet and the diameter is two inches). Examples of low-deflection roller tubes are described in greater detail in U.S. Patent Application Publication No. 2016/0326801, published Nov. 10, 2016, entitled LOW-DEFLECTION ROLLER SHADE TUBE FOR LARGE OPENINGS, the entire disclosure of which is hereby incorporated by reference.
(17) The flexible material **120** may include a first end (e.g., a top or upper end) that is coupled to the roller tube **110** and a second end (e.g., a bottom or lower end) that is coupled to a hembar **140**. The hembar **140** may be configured, for example weighted, to cause the flexible material **120** to

hang vertically. Rotation of the roller tube **110** may cause the hembar **140** to move toward or away from the roller tube **110** between the raised and lowered positions.

(18) The flexible material **120** may be any suitable material, or form any combination of materials. For example, the flexible material **120** may be "scrim," woven cloth, non-woven material, light-control film, screen, and/or mesh. The motorized window treatment **100** may be any type of window treatment. For example, the motorized window treatment **100** may be a roller shade as illustrated, a soft sheer shade, a drapery, a cellular shade, a Roman shade, or a Venetian blind. As shown, the flexible material **120** may be a material suitable for use as a shade fabric, and may be alternatively referred to as a flexible material. The flexible material **120** is not limited to shade fabric. For example, in accordance with an alternative implementation of the motorized window treatment **100** as a retractable projection screen, the flexible material **120** may be a material suitable for displaying images projected onto the flexible material **120**.

(19) The motorized window treatment **100** may include a drive assembly (e.g., such as the motor drive unit **590** shown in FIG. **3**A). The drive assembly may at least partially be disposed within the roller tube **110**. For example, the drive assembly may be retained within a motor drive unit housing (e.g., such as the motor drive unit housing **580** shown in FIG. **3**A) that is received within the roller tube **110**. The drive assembly may include a control circuit that may include a microprocessor and may be mounted to a printed circuit board. The drive assembly may be powered by a power source (e.g., an alternating-current or direct-current power source) provided by electrical wiring and/or batteries (e.g., as shown in FIGS. **3**A-**5**). The drive assembly may be operably coupled to the roller tube **110** such that when the drive assembly is actuated, the roller tube **110** rotates. The drive assembly may be configured to rotate the roller tube **110** of the example motorized window treatment **100** such that the flexible material **120** is operable between the raised position and the lowered position. The drive assembly may be configured to rotate the roller tube **110** while reducing noise generated by the drive assembly (e.g., noise generated by one or more gear stages of the drive assembly). Examples of drive assemblies for motorized window treatments are described in greater detail in commonly-assigned U.S. Pat. No. 6,497,267, issued Dec. 24, 2002, entitled MOTORIZED WINDOW SHADE WITH ULTRAQUIET MOTOR DRIVE AND ESD PROTECTION, and U.S. Pat. No. 9,598,901, issued Mar. 21, 2017, entitled QUIET MOTORIZED WINDOW TREATMENT SYSTEM, the entire disclosures of which are hereby incorporated by reference.

(20) The motorized window treatment **100** may be configured to enable access to one or more ends of the roller tube **110** while remaining secured to the mounting brackets **130**A, **130**B. For example, the motorized window treatment **100** may be adjusted (e.g., pivoted or slid) between an operating position (e.g., as shown in FIG. **1**) to an extended position (e.g., as shown in FIG. **1**B) while secured to the mounting brackets **130**A, **130**B. The operating position may be defined as a position in which the roller tube **110** is supported by and aligned with both mounting brackets **130**A, **130**B. The extended position may be defined as a position in which one or more ends of the roller tube **110** are accessible while still attached to the brackets **130**A, **130**B. Operation of the motorized window treatment **100** may be disabled when it is adjusted between the operating position and the extended position. For example, operation of the motorized window treatment **100** may disabled when the extended position is reached. Alternatively, operation of the motorized window treatment **100** may be disabled at some point between the operating position and the extended position, for example, when the motorized window treatment **100** exits the operating position. Operation of the motorized window treatment **100** may be enabled when it enters the operating position.

(21) When in the extended position, the one or more ends of the roller tube **110** may be accessed, for example, to replace batteries, adjust one or more settings, make an electrical connection, repair one or more components, and/or the like. One or more of the mounting brackets **130**A, **130**B may enable an end of the roller tube **110** to be accessed when the motorized window treatment is in the extended position. One or more of the mounting brackets **130**A, **130**B may include a sliding

portion to enable the end of the roller tube **110** to be accessible. For example, a first portion (e.g., sliding portion) of one or more of the mounting brackets **130**A, **130**B may extend from a second portion (e.g., fixed portion). For example, a sliding portion of one or more of the mounting brackets **130**A, **130**B may be adjusted with respect to a fixed portion, for example, to expose a respective end of the roller tube **110**.

(22) One end of the roller tube may slide out when the motorized window treatment is in the extended position. For example, one of the mounting brackets (e.g., mounting bracket **130**A) may be configured to slide out and the other one of the mounting brackets (e.g., mounting bracket **130**B) may remain stationary when the motorized window treatment **100** (e.g., the roller tube **110**) is in the extended position, for example, as shown in FIG. **1**B. The extended position of the motorized window treatment **100** may include a first end **112** of the roller tube **110** proximate to a first mounting bracket (e.g., mounting bracket **130**A) being further from a window and/or the structure to which the first mounting bracket is anchored than when the motorized window treatment **100** is in the operating position. A second end **114** (e.g., opposite the first end **112**) of the roller tube **110** proximate to the second mounting bracket (e.g., mounting bracket **130**B) may remain substantially fixed when the motorized window treatment **100** is in the extended position, for example, as shown in FIG. **1**B. Stated differently, the roller tube **110** may pivot between the operating position and the extended position. The second end **114** of the roller tube **110** and the mounting bracket **130**B may define a fulcrum about which the motorized window treatment **100** (e.g., the roller tube **110**) pivots.

(23) Alternatively, both ends of the roller tube may slide out when the motorized window treatment is in the extended position. For example, both of the mounting brackets **130**A, **130**B may be configured to slide out. That is, both of the mounting brackets **130**A, **130**B may include sliding portions. In this configuration, both the first end **112** and the second end **114** may be further from the window and/or the structure when the motorized window treatment **100** is in the extended position. Stated differently, the motorized window treatment **100** may slide between the operating position and the extended position. When both ends of the roller tube are configured to slide out, two people may be required to operate the motorized window treatment **100** between the operating position and the extended position.

(24) When the motorized window treatment **100** is in the extended position, a motor drive unit housing end **150** (e.g., cap **250** shown in FIGS. **2**A and **2**B) may be exposed (e.g., accessible). The motor drive unit housing end **150** may be located proximate to the first end **112** of the roller tube **110**. The motor drive unit housing end **150** may cover a cavity of the roller tube **110**. The motor drive unit housing end **150** may be configured to be removably secured to the roller tube **110** (e.g., the first end **112** of the roller tube **110**). For example, the motor drive unit housing end **150** may be configured to be secured within the cavity. The motor drive unit housing end **150** may be configured to retain one or more components (e.g., such as the batteries **260** shown in FIGS. **2**A and **2**B).

(25) The motor drive unit housing end **150** may include a control button **152**. The control button **152** may be backlit. For example, the control button **152** may include a light pipe (e.g., may be translucent or transparent) that is illuminated by a light emitting diode (LED) within the motor drive unit housing. The control button **152** may be configured to enable a user to change one or more settings of the motorized window treatment **100**. For example, the control button **152** may be configured to change one or more wireless communication settings and/or one or more drive settings. The control button **152** may be configured to enable a user to pair the motorized window treatment **100** with a remote control device to allow for wireless communication between the remote control device and a wireless communication circuit (e.g., an RF transceiver) in the motor drive unit housing end **150**. The control button **152** may be configured to provide a status indication to a user. For example, the control button **152** may be configured to flash and/or change colors to provide the status indication to the user. The status indication may indicate when the motorized window treatment **100** is in a programming mode.

(26) The motor drive unit housing end **150** may include a disable actuator **154** that is configured to deactivate (e.g., automatically deactivate) the drive assembly when the roller tube **110** is not in the operating position. For example, the disable actuator **154** may be configured to disable the drive assembly such that the covering material cannot be raised or lowered when the roller tube **110** is not in the operating position. The disable actuator **154** may disable a motor of the drive assembly, for example, when the roller tube **110** is pivoted (e.g., or slid) from the operating position to the extended position. The disable actuator **154** may enable the motor when the roller tube **110** reaches the operating position. The disable actuator **154** may be a button, a magnetic sensor, an IR sensor, a switch, and/or the like.

(27) FIGS. **2**A and **2**B depict an example battery-powered motorized window treatment **200** (e.g., such as the motorized window treatment **100** shown in FIG. **1**). The battery-powered motorized window treatment **200** may include a roller tube **210** (e.g., such the roller tube **110** shown in FIG. **1**), a flexible material **120** (e.g., a covering material) windingly attached to the roller tube **210**, a drive assembly (e.g., such as the motor drive unit **590** shown in FIG. **3**A), and a plurality of batteries **260**. The battery-powered motorized window treatment **200** may further include a hembar **240** (e.g., such as the hembar **140** shown in FIGS. **1**A and **1**B) and one or more mounting brackets **230**A, **230**B (e.g., such as the mounting brackets **130**A, **130**B shown in FIGS. **1**A and **1**B). The battery-powered motorized window treatment **200** (e.g., the drive assembly) may be powered by the batteries **260**. Although the battery-powered motorized window treatment **200** is shown with four batteries **260**, it should be appreciated that the battery-powered motorized window treatment **200** may include a greater or smaller number of batteries. The roller tube **210** may define a longitudinal axis **216**. The longitudinal axis **216** may extend along a longitudinal direction L.

(28) The battery-powered motorized window treatment **200** may include a cap **250** that is configured to retain the batteries **260** within the roller tube **210**. The cap **250** may define an outer surface **252** with a button **254**. The button **254** may be backlit. For example, the button **254** may include a light pipe that is illuminated by an LED within the cap **250**. The button **254** may be configured to enable a user to change one or more settings of the battery-powered motorized window treatment **200**. The button **254** may be configured to enable a user to pair the battery-powered motorized window treatment **200** with a remote control device to allow for wireless communication between the remote control device and the wireless communication circuit in the cap **250**. The button **254** may be configured to provide a status indication to a user. For example, the button **254** may be configured to flash and/or change colors to provide the status indication to the user. The button **254** may indicate when the battery-powered motorized window treatment **200** is in a programming mode, for example, via the status indication.

(29) The drive assembly may be at least partially received within the roller tube **210**. For example, the roller tube **210** may define a cavity **211** (e.g., a battery compartment) that is configured to receive one or more components of the drive assembly. The cavity **211** may be defined by the inner surface **213** of the roller tube **210**. The cavity **211** may be accessible when the battery-powered motorized window treatment **200** is in the extended position (e.g., pivoted) and the cap **250** is removed.

(30) The battery-powered motorized window treatment **200** may include a battery holder **270**. The battery holder **270** may be configured to keep the batteries **260** fixed in place securely while the batteries **270** are providing power to the drive assembly. The battery holder **270** may be configured to clamp the batteries **260** together (e.g., as shown in FIG. **2**A) such that the batteries **260** can be removed from the battery-powered motorized window treatment **200** at the same time (e.g., together). The battery holder **270** may include a head **272**, a base **274**, and an arm **276** connecting the head **272** and the base **274**. The battery holder **270** may create a spring tension to hold the batteries **260** together. For example, the head **272**, the base, **274**, and the arm **276** may be configured to apply a tension force to the batteries **260**.

(31) The head **272** may define an aperture **273** that is configured to receive a nub **263** of one of the

batteries **260**, for example, such that the nub **263** can be electrically connected to the cap **250**. For example, the nub **263** may extend beyond the head **272** when the batteries are clamped within the battery holder **270**. The base **274** may define an aperture configured to receive a spring (e.g., such as spring **486** shown in FIG. **3**A) to electrically connect the batteries **260** to a printed circuit board of the motor drive unit. For example, the spring may be located within the cavity **211** proximate to the motor drive unit. Additionally or alternatively, the base **274** may include an electrical contact (e.g., a negative contact). The electrical contact of the battery holder **270** may be electrically connected to the printed circuit board of the motor drive unit. The base **274** (e.g., the electrical contact) may be configured to abut the spring within the roller tube **210** (e.g., the motor drive unit housing). One or more of the batteries **260** may be received (e.g., at least partially received) within the base **274**. The battery holder **270** may be configured to be removed from the roller tube **210** (e.g., the cavity **211** of the roller tube **210**) while clamping the batteries **260**. Although the battery holder **270** is shown having the arm **276**, it should be appreciated that the battery holder **270** may include alternate means for clamping and/or securing the batteries **260** together. For example, the battery holder **270** may include a sleeve between the head **272** and the base **274**. The sleeve may be configured to surround the batteries **260**.

(32) The battery holder **270** may be configured to be removed (e.g., completely removed as shown in FIG. **2**A) from the roller tube **210**. When the battery holder **270** is removed from the roller tube **210**, the batteries **260** may be removed from the battery holder **270** (e.g., as shown in FIG. **2**B) while still clamped together. Replacement batteries may be installed in the battery holder **270** and the battery holder **270** may be installed within the cavity **211** of the roller tube **210**. When the battery holder **270** is installed within the roller tube **210** (e.g., the cavity **211**), the cap **250** may be removably secured to the roller tube **210** (e.g., the end **212**), for example, to secure the battery holder **270** within the roller tube **210**. Additionally or alternatively, the cap **250** may be configured to be removably secured to the motor drive unit housing.

(33) FIG. **3**A depicts an example battery-powered motorized window treatment **500** (e.g., such as the motorized window treatment **100** shown in FIGS. **1**A and **1**B, and/or the battery-powered motorized window treatment **200** shown in FIGS. **2**A and **2**B) in an operating position. The battery-powered motorized window treatment **500** may include a roller tube **510**, a motor drive unit **590**, a plurality of batteries **560**, and one or more mounting brackets **530**. The operating position may be defined as a position in which the roller tube **510** is supported by and aligned with both mounting brackets **530**. The battery-powered motorized window treatment **500** may be configured to be operated between the operating position and an extended position, for example, to enable access to replace the batteries **560**. The extended position may be defined as a position in which one or more ends of the roller tube **510** are accessible while still attached to the mounting brackets **530**. The extended position may define a pivoted position, for example, as shown in FIG. **2**, where one of the mounting brackets **530** extends such that the batteries **560** are accessible via the end of the roller tube **510**. Although not shown in FIG. **3**A, the battery-powered motorized window treatment **500** may include a flexible material windingly attached to the roller tube **510** and a hembar that is coupled to a bottom or lower end of the flexible material.

(34) The mounting bracket **530** may be configured to attach the battery-powered motorized window treatment **500** to a horizontal structure (e.g., such as a ceiling). The mounting bracket **530** may define a base **538** and an arm **532**. The base **538** and the arm **532** may define a stationary portion of the mounting bracket **530**. The mounting bracket **530** may define a translating portion **534**. The translating portion **534** may include an attachment member **533** that is configured to receive an end of the roller tube **510** and/or a motor drive unit housing **580**. The attachment member **533** may define an aperture. The base **538** may be configured to attach the mounting bracket **530** to a structure. The structure may include a window frame, a wall, a ceiling, or other structure, such that the motorized window treatment is mounted proximate to an opening (e.g., over the opening or in the opening), such as a window for example. When the mounting bracket **530** is

attached to a vertical structure, such as a wall, the arm **532** of the mounting bracket **530** may extend horizontally (e.g., in the radial direction R) from the base **538**.

(35) The translating portion **534** may be configured to translate the roller tube **510** between the operating position and the extended position. The translating portion **534** may be proximate to the base **538** when in the operating position and distal from the base **538** when in the extended position. The end of the roller tube **510** and/or the motor drive unit housing **580** may be accessible via the aperture (e.g., to replace the batteries **560**) when the translating portion **534** is in the extended position.

(36) The arm **532** may define one or more features that enable the translating portion **534** to be translated between the operating position and the extended position while remaining attached thereto. The translating portion **534** may define one or more corresponding features that are configured to cooperate with the one or more features on the arm **532**. The arm **532** may define one or more slides **535** (e.g., an upper slide and a lower slide). The slides **535** may protrude from an inner surface of the arm **532**. The translating portion **534** may define one or more channels (e.g., an upper channel and a lower channel) that are configured to receive the slides **535**. The translating portion **534** may define a middle slide **536**, for example, between the channels. The arm **532** may define a channel (e.g., a middle channel) that is configured to receive the middle slide **536**. The slides **535**, **536** and the channels may define angled edges (e.g., tapered edges) such that the attachment of the translating position **534** to the arm **532** defines an interlocking slide, e.g., such as a dovetail slide. The translating portion **534** may translate along the slides **535** between the operating position and the extended position. For example, the translating portion **534** may translate along the slides **535** in the radial direction R.

(37) The mounting bracket **530** may be configured to be secured (e.g., locked) in the operating position and the extended position. The mounting bracket **530** (e.g., the translating portion) may define a locking tab. In addition, the mounting bracket **530** may comprise a release button (not shown) that may need to be actuated by a user in order to be released the mounting bracket **530** from the operating position and be moved into the extended position.

(38) The motor drive unit **590** may include a motor drive printed circuit board **592**, an intermediate storage device **594**, a motor **596**, and a gear assembly **598**. The intermediate storage device **594** may include one or more capacitors (e.g., super capacitors) and/or one or more rechargeable batteries. The motor drive unit **590** may be operatively coupled to the roller tube **510**, for example, via a coupler **595**. The coupler **595** may be an output gear that is driven by the motor **596** and transfers rotation of the motor **596** to the roller tube **510**. For example, the coupler **595** may define a plurality of grooves **597** about its periphery. An inner surface of the roller tube **510** may be splined. That is, the inner surface of the roller tube **510** may define a plurality of splines **512**. The grooves **597** may be configured to engage respective splines **512** such that rotation of the motor **596** is transferred to the roller tube **510**, for example, via the coupler **595**. The motor drive unit **590** may be configured to detect when one or more batteries **560** are not installed, for example, in the operating position. When the motor drive unit **590** detects that one or more batteries **560** are not installed and the roller tube **510** is in the operating position, the motor drive unit **590** may prevent rotation of the roller tube **510**. In doing so, the motor drive unit **590** may prevent depletion of the intermediate storage device **594**.

(39) The battery-powered motorized window treatment **500** (e.g., the motor drive unit **590**) may include an inner bearing **520** and an outer bearing **540** that are located external to the roller tube **510**. The inner bearing **520** and the outer bearing **540** may be non-metallic (e.g., plastic) sleeve bearings. The inner bearing **520** and the outer bearing **540** may be captured between the roller tube **510** and the mounting bracket **530**. The inner bearing **520** may engage the motor drive unit housing **580**. The inner bearing **520** may be operatively coupled to the motor drive unit housing **580**. For example, the inner bearing **520** may define splines (not shown) that are configured to be received by grooves **588** around the periphery of the motor drive unit housing **580**. The inner bearing **520**

may be press fit onto the motor drive unit housing **580**. The outer bearing **540** may engage the roller tube **510**. The outer bearing **540** may be operatively coupled to the roller tube **510**. The outer bearing **540** may rotate with the roller tube **510**. The outer bearing **540** may be press fit into engagement with the roller tube **510**. For example, the outer bearing **540** may engage the plurality of splines **512** of the roller tube **510**. The inner bearing **520** may remain stationary with the motor drive unit housing **580** as the roller tube **510** rotates. Stated differently, the roller tube **510** and the outer bearing **540** may rotate about the inner bearing **520** and the motor drive unit housing **580**.
(40) The batteries **560** may be configured to be removed from the roller tube **510**, for example, while the motor drive unit housing **580** remains engaged with the mounting brackets **530**. That is, the batteries **560** may be configured to be removed from the roller tube **510** when the battery-powered motorized window treatment **500** is in the pivoted position. An inside diameter of the inner bearing **520** may be greater than an outer diameter of the batteries **560** and/or the battery holder **570**.
(41) The battery-powered motorized window treatment **500** (e.g., the motor drive unit **590**) may include a battery holder **570** and a cap **550**. The battery holder **570** and the cap **550** may keep the batteries **560** fixed in place securely while the batteries **570** are providing power to the motor drive unit **590** and/or the cap **550**. The battery holder **570** may be configured to clamp the batteries **560** together such that the batteries **560** can be removed from the battery-powered motorized window treatment **200** at the same time (e.g., together).
(42) The battery holder **570** may be received in a motor drive unit cavity **588** of the motor drive unit **590**. The motor drive unit cavity **588** may extend in the longitudinal direction L from an end **581** of the motor drive unit **590** (e.g., the motor drive unit housing **580**) to an internal wall **583** of the motor drive unit **590**. The motor drive unit cavity **588** may be open at the end **581**. The motor drive unit **590** may be received within a roller tube cavity **515**. The roller tube cavity **515** may be open proximate to an end of the roller tube **510**. The roller tube cavity **515** may extend in the longitudinal direction L along the entire length of the roller tube **510**. The cap **550** may be configured to cover the end **581** to the motor drive unit cavity **588**. For example, the cap **550** may be received (e.g., at least partially) within the motor drive unit cavity **588**. The cap **550** may include a button **552**, one or more wireless communication components mounted to the control interface printed circuit board **554**, and an electrical contact **556** electrically connected to the control interface printed circuit board **554**. The electrical contact **556** may be a positive electrical contact, for example, as shown in FIG. **3**A. Alternatively, the electrical contact **556** may be a negative electrical contact. The cap **550** may include a switch **555** (e.g., a mechanical tactile switch) mounted to the control interface printed circuit board **554** and configured to be actuated in response to actuations of the button **552**. The button **552** may operate as a light pipe (e.g., may be translucent or transparent), and may be illuminated by an LED (not shown) mounted to the control interface printed circuit board **554**.
(43) The cap **550** may include a switch or button (e.g., button **154** shown in FIG. **1**B) that is configured to deactivate (e.g., automatically deactivate) the motor drive unit **590** when the roller tube **510** is not in the operating position. The switch or button may disable the motor **596** of the motor drive unit **590**, for example, when the roller tube **510** is pivoted (e.g., or slid) from the operating position to the extended position. The switch or button may enable the motor **596** when the roller tube **510** reaches the operating position.
(44) The batteries **560** may be located between the cap **550** (e.g., the wireless communication components of the motor drive unit **590** of the battery-powered motorized window treatment **500**) and the motor drive unit **590**. For example, the wireless communication components in the cap **550** may be located at a first end of the batteries **560** installed in the roller tube **510** and the motor drive unit **590** may be located at an opposed second end of the batteries **560**.
(45) The one or more wireless communication components within the cap **550** may be electrically coupled to an antenna (not shown). The antenna may be a loop antenna that is located around a

periphery of the radio printed circuit board **554**. Alternatively, the antenna may be a monopole. The antenna may be located proximate to a gap **505** between the bracket **530** and the roller tube **510**. The gap **505** includes non-metal components such that radio-frequency interference and/or shielding is minimized. For example, the battery-powered motorized window treatment **500** may not include metal components at the gap **505**. The inner bearing **520** and/or the outer bearing **540** may be disposed within or proximate to the gap **505**.

(46) The gap **505** between the roller tube **510** and the bracket **530** may also be configured to enable a predetermined tolerance (e.g., angular misalignment tolerance) between the roller tube **510** and the bracket **530** in a pivoted position. For example, when the battery-powered motorized window treatment **500** is in the pivoted position, the gap **505** may enable a portion of the roller tube **510** to be closer to the bracket **530** (e.g., without contacting the bracket **530**) than another portion of the roller tube **510**. When the battery-powered motorized window treatment **500** is in the pivoted position, the gap **505** may be configured such that the roller tube **510** does not abut the bracket **530**.

(47) The electrical contact **556** may be electrically connected to the control interface printed circuit board **554**. The button **552** may be backlit. For example, the button **552** may include a light pipe that is illuminated by the LED within the cap **550** and mounted to the control interface printed circuit board **554**. The button **552** may be configured to enable a user to change one or more settings of the battery-powered motorized window treatment **500**. For example the button **552** may be configured to change one or more settings of the control interface printed circuit board **554** and/or a motor printed circuit board **592**. The button **552** may be configured to enable a user to pair the battery-powered motorized window treatment **500** with a remote control device to allow for wireless communication between the remote control device and the wireless communication circuit mounted to the control interface printed circuit board **554** in the cap **250**. The button **552** may be configured to provide a status indication to a user. For example, the control button **552** may be configured to flash and/or change colors to provide the status indication to the user. The button **552** may be configured to indicate (e.g., via the status indication) whether the motor drive unit **590** is in a programming mode.

(48) The control interface printed circuit board **554** and the motor printed circuit board **592** may be electrically connected. For example, the battery-powered motorized window treatment **500** may include a ribbon cable **586**. The ribbon cable **586** may be attached to the control interface printed circuit board **554** and the motor printed circuit board **592**. The ribbon cable **586** may be configured to electrically connect the control interface printed circuit board **554** and the motor printed circuit board **592**. The ribbon cable **586** may terminate at the control interface printed circuit board **554** and the motor printed circuit board **592**. For example, the ribbon cable **586** may extend within the cavity **515**. The ribbon cable **586** may include electrical conductors for providing power from the batteries **560** to the control interface printed circuit board **554** and/or the motor printed circuit board **592**. The ribbon cable **586** may include electrical conductors for conducting control signals (e.g., for transmitting one or more messages) between the control interface printed circuit board **554** and the motor printed circuit board **596**. For example, the ribbon cable **586** may be configured to conduct power and/or control signals between the control interface printed circuit board **554** and the motor printed circuit board **592**.

(49) FIG. **3**B is a side view of an idler end of the example battery-powered motorized window treatment **500**. The battery-powered motorized window treatment **500** may include an idler shaft **514** and an idler coupler **543**. The idler shaft **514** may be configured to support an idler end **511** of the battery-powered motorized window treatment. The idler shaft **514** may remain stationary as the roller tube **510** rotates. The battery-powered motorized window treatment **500** may include idler bearings **544**. The idler bearings **544** may be configured to support the roller tube **510** while enabling the roller tube **510** to rotate about the idler shaft **514**. The idler coupler **543** may be configured to operatively couple the roller tube **510** to the idler bearings **544**.

(50) The battery-powered motorized window treatment **500** may include a spring assist assembly

516 (e.g., a torsion spring assembly). The spring assist assembly **516** may include a spring **517** (e.g., a torsion spring), a bracket coupling portion **518**, and a roller tube coupling portion **508**. The bracket coupling portion **518** may be attached to the idler shaft **514** (e.g., the idler arm **513**) such that the bracket coupling portion **518** remains stationary as the roller tube **510** rotates. The roller tube coupling portion **508** may be operatively coupled to the roller tube **510** (e.g., the splines **512**) such that the roller tube coupling portion **508** rotates with the roller tube **510**. The spring **517** may be attached to the bracket coupling portion **518** at one end and to the roller tube coupling portion **508** at the other end. The spring **517** may be configured to coil and uncoil as the roller tube **510** rotates (e.g., depending on the direction of rotation). For example, the torque applied by the spring **517** to the roller tube **510** may change as the roller tube rotates.
(51) The spring assist assembly **516** may be configured to assist the motor drive unit **590** to operate the battery-powered motorized window treatment **500**. For example, the spring assist assembly **516** may reduce the torque required from the motor drive unit **590** to raise and/or lower the covering material of the battery-powered motorized window treatment **500**. The spring assist assembly **516** may prolong the life of the batteries **560**, for example, by assisting the motor drive unit **590**. The spring assist assembly **516** may be coupled to the roller tube **510** for providing a constant torque on the roller tube **510** in a direction opposite a direction of the torque provided on the roller tube **510** by the motor drive unit **590**. For example, the spring assist assembly **516** may provide a torque on the roller tube **510** opposite a torque provided by the motor drive unit **590** to raise the covering material to a position approximately midway between the fully-lowered and fully-raised position without substantial energy being provided by the motor unit **590**. The torque applied by the spring assist assembly **516** on the roller tube **510** may increase as the covering material is lowered. This increasing torque applied by the spring assist assembly **516** may balance the increasing torque created be more of the covering material hanging from the roller tube **510**. The balance between the torque applied by the spring assist assembly **516** and the torque applied by the covering material may result in a substantially constant torque on the motor drive unit **590**. For example, the spring assist assembly **516** may be configured such that the motor drive unit **590** can operate at a substantially constant torque as the covering material is raised and lowered (e.g., operated between a raised position and a lowered position).
(52) The spring assist assembly **516** may assist the motor drive unit **590** when raising the covering material above the midway position to the fully-raised position, and the spring assist assembly **516** may provide a torque on the drive shaft resisting downward motion of the covering material when the covering material is lowered from the fully-raised position to the fully-lowered position. The motor drive unit **590** may provide a torque that is configured to wind up the spring assist assembly **516** when the covering material is lowered from the midway position to the fully-lowered position.
(53) When the roller tube **510** is mounted using two brackets **530**, **531** that may be translated into the extended position, the spring assist assembly **516** may be configured to be adjusted (e.g., pre-wound) at the installation site (e.g., when the roller tube **510** is mounted to the mounting brackets **530**, **531**). Pre-winding the spring assist assembly **516** may enable the spring assembly **516** to provide a constant torque on the roller tube **510** during operation of the motor drive unit **590**. Pre-winding the spring assist assembly **516** at the installation site may eliminate the need to pre-wind the spring assist assembly **516** during manufacturing (e.g., at the factory). Pre-winding the spring assist assembly **516** during manufacturing may result in pre-winding in the wrong direction, too many turns, and/or not enough turns for the application. Pre-winding the spring assist assembly **516** during manufacturing may require a locking mechanism to hold the roller tube **510** such that the pre-wound spring assist assembly **516** does not unwind. Pre-winding the spring assist assembly **516** at the installation site may allow for more precise pre-winding settings, eliminate factory pre-winding errors, and eliminate the need for a locking mechanism to hold the roller tube **510** in place prior to installation.
(54) Motorized window treatments tend to be operated intermittently throughout the day.

Motorized window treatments may draw high peak currents for a short amount of time to drive the motor to move the position of the covering material, followed by long periods of nearly zero current whilst the shade is stationary. As such, motorized window treatments can be considered “peaky loads”—loads that draw high peaks of current for relatively short periods of time and relatively infrequently throughout the day. Further, some motorized window treatments use batteries (e.g., traditional alkaline batteries) as a power source to power the motor that moves the covering material. Typically, the motors of such motorized window treatments are driven directly from the battery voltage of the batteries. While the motor is being driven to move the covering material of the motorized window treatment, the motor draws a large amount of current for a short duration of time from the batteries thus causing the amount of energy stored within the batteries to decrease. However, there is a non-linear relationship between the current drawn (e.g., the peak current drawn) from the batteries and the amount of energy (Joules) available from the batteries (e.g., and the expected lifetime of the batteries). For instance, if batteries supply power at a relatively high peak current, the total energy level of the batteries may decrease faster and the battery lifetime may be shorter than if system is configured to supply power from the batteries at a relatively low current (e.g., the batteries may drain faster when the current drawn is higher, and the batteries may drain slower when the current drawn is lower). Accordingly, directly driving the motor of a motorized window treatment from the batteries with high peak currents, as in prior art motorized window treatments, may be suboptimal for the lifetime performance of the batteries.

(55) Further, infrastructure and existing devices are traditionally designed to handle the peak power conditions. But, in instances where the load is one that draws high peaks of current for relatively short periods of time and relatively infrequently throughout the day (e.g., a “peaky load”), the infrastructure and/or devices that support such loads may sit idle for the vast majority of the day (e.g., over 90% of the time). This results in a higher overall cost to manufacture and maintain the system since the infrastructure and/or devices are designed to handle the peak power conditions, which are infrequently required.

(56) The motorized window treatment described herein may be configured to decouple the power and time of the infrequent, high peak current demands of such loads. For example, the motorized window treatments described herein may be configured to cause the batteries to supply power at a relatively small current for a long duration of time. The motorized window treatments may include an internal energy storage element (e.g., one or more supercapacitors, one or more rechargeable batteries, and/or one or more lithium iron phosphate batteries) and circuitry for limiting the current drawn from the batteries, such that a small constant current (e.g. a desired average current) may be drawn from the batteries over a long period of time to prolong the lifetime of the batteries. For example, the motorized window treatments described herein may be configured to reduce the peak current drawn from the batteries, for example, by charging the internal energy storage element slowly over time (e.g., thereby reducing the peak current drawn over a short period of time from the batteries) and then powering the load (e.g., motor) using the energy stored within the internal energy storage element. Accordingly, the motorized window treatments described herein may utilize the internal energy storage element to draw a small constant current from the batteries over a long period of time to extend the lifetime (e.g., and increase the total energy output) of the batteries, reduce the peak current and/or voltage draws from the batteries, and/or reduce the likelihood of battery failures.

(57) The batteries may be referred to as primary batteries, while the energy storage element may be referred to as a secondary battery. The primary batteries may be replaceable by the user, for example, when they fall below a threshold energy level. The primary batteries may, for example, be alkaline batteries, such as those that are available off-the-shelf. The energy storage element, or secondary battery, may be a semi-permanent power source of the device. For example, the energy storage element may be a commercial power source, such as one or more lithium iron phosphate batteries or supercapacitors. The energy storage element may be integrated into the device and not

intended for the user to remove or replace (e.g., outside of rare exceptions where, for example, a technician may service the device). As noted, the primary batteries may be comprised of a different battery chemistry than that of the energy storage element. Further, in some examples, the batteries may have an internal resistance that is multiple factors (e.g., 10 times) greater than the internal resistance of the energy storage element. As an example, the batteries may be characterized by a voltage drop of at least 0.2 V in response to a draw of 0.5 W, and the energy storage element may be characterized by a voltage drop of no more than 0.04 V in response to a draw of 0.5 W. Further, in some examples, the batteries the batteries (e.g., each battery) may be characterized by a voltage drop in response to a draw of 0.5 W that is at least 10 times larger than a voltage drop of the energy storage element in response to a draw of 0.5 W.

(58) FIG. **4** is a block diagram of an example motor drive unit **600** of a motorized window treatment (e.g., the motor drive unit **590** of the motorized window treatment **500** of FIG. **3**A). The motor drive unit **600** may comprise a motor **610** (e.g., a direct-current (DC) motor) that may be coupled for raising and lowering a covering material. For example, the motor **610** may be coupled to a roller tube **510** of the motorized window treatment for rotating the roller tube for raising and lowering a flexible material (e.g., a shade fabric). The motor drive unit **600** may comprise a load control circuit, such as a motor drive circuit **620** (e.g., an H-bridge drive circuit) that may generate a pulse-width modulated (PWM) voltage V.sub.PWM for driving the motor **610** (e.g., to move the covering material between a fully-raised and fully-lowered position). In addition, the control circuit **630** may be configured to generate a direction signal for controlling the direction of rotation of the motor **610**.

(59) The motor drive unit **600** may comprise a control circuit **630** for controlling the operation of the motor **610**. The control circuit **630** may comprise, for example, a microprocessor, a programmable logic device (PLD), a microcontroller, an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any suitable processing device or control circuit. The control circuit **630** may be configured to generate a drive signal V.sub.DRV for controlling the motor drive circuit **620** to control the rotational speed of the motor **610** (e.g. the motor drive circuit **620** receives the drive signal V.sub.DRV and controls, for example, an H-bridge circuit with appropriate PWM signals in response to the drive signal). In examples, the drive signal VDRV may comprise a pulse-width modulated signal, and the rotational speed of the motor **610** may be dependent upon a duty cycle of the pulse-width modulated signal. In examples, the control circuit **630** may directly control the motor **610** (e.g. in a configuration with no separate motor drive circuit **620**). For example, the control circuit may generate two PWM signals for controlling the duty cycle and the polarity (e.g. controlling the speed and direction) of the motor **610**. The control circuit **630** may be configured to generate a direction signal V.sub.DIR for controlling the motor drive circuit **620** to control the direction of rotation of the motor **610**. The control circuit **630** may be configured to control the motor **610** to adjust a present position P.sub.PRES of the covering material of the motorized window treatment between a fully-raised position P.sub.RAISED and a fully-lowered position P.sub.LOWERED.

(60) The motor drive unit **600** may include a rotational sensing circuit **640**, e.g., a magnetic sensing circuit, such as a Hall effect sensor (HES) circuit, which may be configured to generate two signals V.sub.S1, V.sub.S2 (e.g., Hall effect sensor signals) that may indicate the rotational position and direction of rotation of the motor **610**. The rotational sensing circuit **640** (e.g., HES circuit) may comprise two internal sensing circuits for generating the respective signals V.sub.S1, V.sub.S2 (e.g., HES signals) in response to a magnet that may be attached to a drive shaft of the motor. The magnet may be a circular magnet having alternating north and south pole regions, for example. For example, the magnet may have two opposing north poles and two opposing south poles, such that each sensing circuit of the rotational sensing circuit **640** is passed by two north poles and two south poles during a full rotation of the drive shaft of the motor. Each sensing circuit of the rotational sensing circuit **640** may drive the respective signal V.sub.S1, V.sub.S2 to a high state when the

sensing circuit is near a north pole of the magnet and to a low state when the sensing circuit is near a south pole. The control circuit **630** may be configured to determine that the motor **610** is rotating in response to the signals V.sub.S1, V.sub.S2 generated by the rotational sensing circuit **640**. In addition, the control circuit **630** may be configured to determine the rotational position and direction of rotation of the motor **610** in response to the signals V.sub.S1, V.sub.S2.

(61) The motor drive unit **600** may include a communication circuit **642** that may allow the control circuit **630** to transmit and receive communication signals, e.g., wired communication signals and/or wireless communication signals, such as radio-frequency (RF) signals. For example, the motor drive unit **600** may be configured to communicate messages (e.g., digital messages) with external control devices (e.g., other motor drive units). The communication circuit **642** may be internal to a housing of the motor drive unit **600**. The motor drive unit **600** may also, or alternatively, be coupled to an external RF communication circuit (e.g., located outside of the motor drive unit) for transmitting and/or receiving the RF signals.

(62) The motor drive unit **600** may communicate with one or more input devices, e.g., such as a remote control device, an occupancy sensor, a daylight sensor, and/or a shadow sensor. The remote control device, the occupancy sensor, the daylight sensor, and/or the shadow sensor may be wireless control devices (e.g., RF transmitters) configured to transmit messages to the motor drive unit **600** via the RF signals. For example, the remote control device may be configured to transmit digital messages via the RF signals in response to an actuation of one or more buttons of the remote control device. The occupancy sensor may be configured to transmit messages via the RF signals in response to detection of occupancy and/or vacancy conditions in the space in which the motorized window treatment is installed. The daylight sensor may be configured to transmit digital messages via RF signals in response to a measured amount of light inside of the space in which the motorized window treatment is installed. The shadow sensor may be configured to transmit messages via the RF signals in response to detection of a glare condition outside the space in which the motorized window treatment is installed.

(63) The motorized window treatment may be configured to control the covering material according to a timeclock schedule. The timeclock schedule may be stored in memory of the motor drive unit **600**. The timeclock schedule may be defined by a user (e.g., a system administrated through a programming mode). The timeclock schedule may include a number of timeclock events. The timeclock events may have an event time and a corresponding command or preset. The motorized window treatment may be configured to keep track of the present time and/or day. The motorized window treatment may transmit the appropriate command or preset at the respective event time of each timeclock event.

(64) The motor drive unit **600** may further comprise a user interface **644** having one or more actuators (e.g., mechanical switches) that allow a user to provide inputs to the control circuit **630** during setup and configuration of the motorized window treatment (e.g., in response to actuations of one or more buttons (e.g., the control button **152**). The control circuit **630** may be configured to control the motor **610** to control the movement of the covering material in response to a shade movement command received from the communication signals received via the communication circuit **642** or the user inputs from the buttons of the user interface **644**. The control circuit **620** may be configured to enable (e.g., via the control button **152** and/or the user interface **644**) a user to pair the motorized window treatment with a remote control device and/or other external devices to allow for wireless communication between the remote control device and/or other external devices and the communication circuit **642** (e.g., an RF transceiver). The user interface **644** (e.g., the control button **152**) may be configured to provide a status indication to a user. For example, user interface **644** (e.g., the control button **152**) may be configured to flash and/or change colors to provide the status indication to the user. The status indication may indicate when the motorized window treatment is in a programming mode. The user interface **644** may also comprise a visual display, e.g., one or more light-emitting diodes (LEDs), which may be illuminated by the control

circuit **630** to provide feedback to the user of the motorized window treatment system.
(65) The motor drive unit **600** may comprise a memory **646** configured to store the present position P.sub.PRES of the covering material and/or the limits (e.g., the fully-raised position P.sub.RAISED and the fully-lowered position P.sub.LOWERED), association information for associations with other devices and/or instructions for controlling the motorized window treatment. The memory **646** may be implemented as an external integrated circuit (IC) or as an internal circuit of the control circuit **630**. The memory **646** may comprise a computer-readable storage media or machine-readable storage media that maintains computer-executable instructions for performing one or more as described herein. For example, the memory **646** may comprise computer-executable instructions or machine-readable instructions that include one or more portions of the procedures described herein. The control circuit **630** may access the instructions from memory **646** for being executed to cause the control circuit **230** to operate as described herein, or to operate one or more other devices as described herein. The memory **646** may comprise computer-executable instructions for executing configuration software. The computer-executable instructions may be executed to perform procedures **700**, **800**, **900**, **1000**, and/or **1100** as described herein. Further, the memory **646** may have stored thereon one or more settings and/or control parameters associated with the motor drive unit **600**.
(66) The motor drive unit **600** may comprise a compartment **664** (e.g., which may be an example of the battery compartment **211** of the window treatment) that is configured to receive a DC power source. In some examples, the compartment **664** may be internal to the motor drive unit **600**. In other examples, the compartment **664** may be external to the motor drive unit **600**. In the example shown in FIG. **4**, the DC power source is one or more batteries **660**. In examples, one or more alternate DC power sources may be coupled in parallel with the one or more batteries **660**, or in some examples be used as an alternative to the batteries **660**. For example, the alternative DC power sources may comprise one or more of a solar energy receiving circuit (e.g., a solar cell and/or a photovoltaic cell), an ultrasonic energy receiving circuit, and/or a radio-frequency (RF) energy receiving circuit, and other suitable energy harvesting circuits. The alternate DC power source may be used to perform the same and/or similar functions as the one or more batteries **660**. The DC power source may be characterized by a larger equivalent series resistance than the energy storage element **654**.
(67) In the illustrated example, the compartment **664** may be configured to receive one or more batteries **660** (e.g. four “D” batteries), such as the batteries **260**, **560** of FIGS. **2**A, **2**B, **3**. The batteries **660** may provide a battery voltage V.sub.BATT to the motor drive unit **600**. The batteries **660** may be referred to as primary batteries. The batteries **660** may be replaceable by the user, for example, when they fall below a threshold energy level. The batteries **660** may, for example, be alkaline batteries, such as those that are available off-the-shelf.
(68) The motor drive unit **600** may comprise a first filter circuit **670**, a current limiting circuit, such as a power converter circuit **652**, and an energy storage element **654** (e.g., an intermediate energy storage element). In some examples, the motor drive unit **600** may include a second power converter, such as a boost converter circuit (not shown). Also, in some examples, the second power converter may be omitted from the motor drive unit **600**.
(69) The energy storage element **654** may comprise any combination of one or more supercapacitors, one or more rechargeable batteries, and/or other suitable energy storage devices. In some examples, the energy storage element **654** may be referred to as a secondary battery. The energy storage element **654** may be a semi-permanent power source of the motor drive unit **600**. For example, the energy storage element **654** may be a commercial power source, such as one or more iron phosphate lithium batteries or supercapacitors. The energy storage element **654** may be integrated into the motor drive unit **600** and not intended for the user to remove or replace (e.g., outside of rare exceptions where, for example, a technician may service the motor drive unit **600**). As previously noted, the batteries **660** may be comprised of a different battery chemistry than that

of the energy storage element **654**. Further, in some examples, the batteries **660** may have an internal resistance that is multiple factors (e.g., 10 times) greater than the internal resistance of the energy storage element **654**.

(70) The first filter circuit **670** may receive the battery voltage V.sub.BATT. The power converter circuit **652** may draw a battery current I.sub.BATT from the batteries **660** through the first filter circuit **670**. The first filter circuit **670** may filter high and/or low frequency components of the battery current I.sub.BATT. In some examples, the first filter circuit **670** may be a low-pass filter. Also, in some examples, the first filter circuit **670** may be omitted from the motor drive unit **600**.

(71) The power converter circuit **652** may be configured to limit the current drawn from the batteries **660** (e.g. allowing a small constant current to flow from the batteries **660**). The power converter circuit **652** may receive the battery voltage V.sub.BATT (e.g., V.sub.IN) via the first filter circuit **670**. In some examples, the power converter circuit **652** may comprise a step-down power converter, such as a buck converter. The power converter circuit **652** may be configured to charge the energy storage element **654** from the battery voltage V.sub.BATT to produce a storage voltage V.sub.S across the energy storage element **654** (e.g., approximately 3.5 volts).

(72) The motor drive unit **600** may include a bus capacitor C.sub.BUS that is configured to store a bus voltage V.sub.BUS. The motor drive circuit **620** may be configured to receive the bus voltage V.sub.BUS and conduct a motor current I.sub.MOTOR through the motor **610** for controlling power delivered to the motor **610** to control movement of the covering material. The motor drive circuit **620** may draw current from the bus capacitor C.sub.BUS along with current from the energy storage element **654** (e.g., via the boost converter circuit, in instances where the motor drive unit comprises the boost converter circuit) or current from the batteries **660** to drive the motor **610**. For instance, in some examples, the motor drive circuit **620** may draw current from the bus capacitor C.sub.BUS and the energy storage element **654** to drive the motor, but not the batteries **660**. In such instances, the power converter circuit **652** may be configured to limit the current drawn from the batteries **660**, for example, by charging the energy storage element **654** and drawing current from the energy storage element **654** to drive the motor **610** (e.g., from the storage voltage V.sub.S). In most cases, for instance, the motor drive circuit **620** may drive the motor **610** by drawing current from the energy storage element **654** and not drawing any current directly from the batteries **660** (e.g., directly from the batteries **600** via the inductor L**664**).

(73) The power converter circuit **652** (e.g., the control circuit **630** controlling the power converter circuit **652**) may control the current drawn from the batteries **660** (e.g., the battery current) such that an open-circuit battery voltage of the batteries **660** reduces by no more than a set percentage, for example, reduces by no more than the set percentage when power is delivered to the motor **610** to control movement of the covering material and/or for a period of time immediately before or after the movement of the covering material. The set percentage may be 10%, or preferably 5%, or more preferably 3%. By preventing large voltage drops, the power converter circuit **652** (e.g., and/or the control circuit **630**) can elongate the useful life of the batteries **660** for providing energy to power the motor **610**. Finally, it should be appreciated that, in some examples, the power converter circuit **652** may be omitted for another current limiting circuit, such as in instances where the battery voltage V.sub.BATT is the same as the storage voltage V.sub.S and power conversion (e.g., a step-up or step-down) is not needed to drive the motor.

(74) The motor drive unit **600** may be configured to control when and how the energy storage element **654** charges from the batteries **660**. The control circuit **630** may control when and how the energy storage element **354** charges from the batteries **660** based on the storage voltage V.sub.S of the energy storage element **654**, such as when the storage voltage V.sub.S of the energy storage element **654** falls below a low-side threshold value (e.g., approximately 3.1 volts). For example, the control circuit **630** may be configured to receive a scaled storage voltage V.sub.SS through a scaling circuit **656** (e.g., a resistive divider circuit). The scaling circuit **656** may receive the storage voltage V.sub.S and may generate the scaled storage voltage V.sub.SS. The control circuit **630** may

determine the magnitude of the storage voltage V.sub.S of the energy storage element **654** based on the magnitude of the scaled storage voltage V.sub.SS. When the control circuit **630** determines that the magnitude of the storage voltage V.sub.S of the energy storage element **654** falls below the low-side threshold value, the control circuit **630** may control a charging enable signal V.sub.EN (e.g., drive the charging enable control signal V.sub.EN high) to enable the power converter circuit **652**. When the power converter circuit is enabled, the power converter circuit **652** may be configured to charge the energy storage element **654** (e.g., from the batteries **660**). For example, when the power converter circuit is enabled, a charging session may be active and, in some examples, the control circuit **630** may set a charging flag to indicate that the charging session is active and the power converter circuit is enabled. When the power converter circuit is disabled, the power converter circuit **652** may be configured to cease charging the energy storage element **654** (e.g., from the batteries **660**). Finally, it should be appreciated that in some example, the motor drive unit **600** may be configured such that the energy storage element **654** cannot be charged at the same time that the motor drive unit **600** is controlling the movement of the covering material and/or for a period of time immediately thereafter.

(75) The motor drive unit **600** may utilize the energy storage element **654** to draw a small constant current from the batteries **660** over a long period of time to extend the lifetime (e.g., and increase the total energy output) of the batteries **660**. For example, the motor drive unit **600** (e.g., the power converter circuit **652** and/or the motor drive circuit **620**) may limit the current drawn by the power converter circuit **652**. The motor drive unit **600** may draw current from the batteries **660** that is less than the limit, but not more. Further, as noted, the motor drive unit **600** may control the current drawn from the batteries **660** (e.g., the battery current) such that an open-circuit battery voltage of the batteries **660** reduces by no more than a set percentage (e.g., 10%, 5%, or 3%) during any instance of power draw from the batteries, for example, when power is delivered to the motor **610** to control movement of the covering material and/or for a period of time immediately before or after the movement of the covering material. The batteries **660** may eventually experience a voltage drop that exceeds the set percentage, but may only do so over an extended period of time (e.g., multiple years, such as 5 year, 10 years, or more, based on how often the window treatment is used).

(76) When enabled, the power converter circuit **652** may be configured to conduct an average current I.sub.AVE (e.g., having a magnitude of approximately 15 milliamps) from the batteries **660**. The magnitude of the average current I.sub.AVE may be much smaller than a magnitude of a drive current, such as a motor current I.sub.MOTOR required by the motor drive circuit **620** to rotate the motor **610**. When the motor drive circuit **620** is driving the motor **610**, the magnitude of the storage voltage V.sub.S of the energy storage element **654** may decrease with respect to time. When the motor drive circuit **620** is not driving the motor **610** and the power converter circuit **652** is charging the energy storage element **654**, the magnitude of the storage voltage V.sub.S may increase (e.g., slowly increase). When the storage voltage V.sub.S of the energy storage element **654** falls below a low-side threshold value (e.g., approximately 3.1 V), the control circuit **630** may enable the power converter circuit to begin charging the energy storage element. The storage voltage V.sub.S may fall below the low-side threshold value after powering movements of the covering material, powering low-voltage components, and/or due to leakage currents over time. When the storage voltage V.sub.S of the energy storage element **654** rises above a high-side threshold value (e.g., approximately 3.6 volts), the control circuit **630** may cease driving the charging enable signal V.sub.EN high to disable the power converter circuit **652** and stop the charging of the energy storage element **654** from the batteries **660**.

(77) As noted herein, the motor drive unit **600** may further comprise a boost converter circuit (not shown). When included, the boost converter may receive the storage voltage V.sub.S and generate the bus voltage V.sub.BUS at a boosted magnitude (e.g., approximately 5 volts) for powering the motor **610**. When boost converter is operating, the bus voltage V.sub.BUS may be larger than the

storage voltage V.sub.S. When the control circuit **630** controls the motor drive circuit **620** to rotate the motor **610**, the boost converter circuit may conduct current from the energy storage element **654** to generate the motor voltage V.sub.BUS.

(78) The motor drive unit **600** may comprise a low-voltage power supply **680**. The low-voltage power supply **680** may receive the battery voltage V.sub.BATT. The low-voltage power supply **680** may be configured to produce a low-voltage supply voltage V.sub.CC (e.g., approximately 3.3 volts) for powering low-voltage circuitry of the motor drive unit **600**, such as the user interface **644**, the communication circuit **642**, the memory **646**, and/or the control circuit **630**. Further, in some examples, the low-voltage power supply **680** may be omitted from the motor drive unit **600** (e.g., if the low-voltage circuitry of the motor drive unit **600** is able to be powered directly from the storage voltage V.sub.S). Additionally or alternatively, the motor drive unit **600** may comprise a low-voltage power supply (not shown) that may receive the storage voltage V.sub.S and generate the low voltage V.sub.CC (e.g., approximately 3.3 V) for powering the control circuit **630** and other low-voltage circuitry of the motor drive unit **600**, e.g., the user interface **644**, the communication circuit **642**, the memory **646**, and/or the control circuit **630**.

(79) The motor drive unit **600** may comprise a first switch, such as a first switching circuit **662**, that is coupled between a first power source of the motor drive unit **600** (e.g., the batteries **660**) and the bus capacitor C.sub.BUS (e.g., between the batteries **660** and the motor drive circuit **620**). The control circuit **630** may generate a first switch control signal V.sub.SW1 for rendering the first switching circuit **662** conductive and non-conductive. The motor drive unit **600** may comprise a second switch, such as a second switching circuit **668**, that is coupled between a second power source of the motor drive unit **600** (e.g., the energy storage element **654**) and the bus capacitor C.sub.BUS (e.g., between the energy storage element **654** and the motor drive circuit **620**). The control circuit **630** may generate a second switch control signal V.sub.SW2 for rendering the second switching circuit **668** conductive and non-conductive. In some examples, the first switching circuit **662** and the second switching circuit **668** may each comprise a bidirectional semiconductor switch, such as a field-effect transistor (FET) inside a full-wave rectifier bridge, two FETs in anti-series connection, and/or other types of bidirectional switching circuits.

(80) The control circuit **630** may be configured to control the first and second switching circuits **662**, **668** to control whether the motor drive circuit **620** draws current from the energy storage element **654** or the batteries **660**. For example, the control circuit **630** may be configured to render the second switching circuit **668** conductive and the first switching circuit **662** non-conductive to allow the motor drive circuit **620** to draw current from the energy storage element **654** to control the power delivered to the motor **610**. Further, as described in more detail herein, the control circuit **630** may be configured to render the first switching circuit **662** conductive and the second switching circuit **668** non-conductive to bypass the first filter circuit **670**, the power converter circuit **652**, the energy storage element **654**, and/or the boost converter circuit (when included) to allow the motor drive circuit **620** to draw current directly from the batteries **660** (e.g., when the energy storage element **654** is depleted). Finally, in some examples, when the motor drive unit **600** is not controlling movement of the covering material (e.g., the motor drive circuit **620** is not drawing current), the control circuit **630** may render the first and second switching circuits **662**, **668** non-conductive. The control circuit **630** may be configured to render (e.g., only render) one of the first switching circuit **662** or the second switching circuit **668** conductive at any given time. Further, in some examples, the motor drive unit **600** may comprise a lock (e.g., a hardware interlock circuit) that is configured to prevent both the first and second switching circuits **662**, **668** from being rendered conductive at the same time. For instance, when included, the hardware interlock circuit may be coupled between the first switch control signal V.sub.SW1 and the second switch control signal V.sub.SW2 to prevent both the first and second switching circuits **662**, **668** from being turned on at the same time.

(81) The motor drive unit **600** may also include a second filter circuit, such as an inductor **664**,

coupled in series between the first switching circuit **662** and the bus capacitor C.sub.BUS. The inductor **664** may be configured to filter the motor current I.sub.MOTOR conducted through the batteries **660** when the first switching circuit **662** is conductive and the motor drive circuit **620** is controlling the power delivered to the motor **610**. Since the motor drive circuit **620** is driving the motor **610** with the PWM voltage V.sub.PWM, the motor current I.sub.MOTOR conducted through the motor **610** may be peaky (e.g., may also be pulse-width modulated). For instance, the inductor **664** may be configured to filter the motor current I.sub.MOTOR such that the battery current I.sub.BATT conducted through the batteries **660** has a substantially DC magnitude. For example, the inductor **664** may filter out high (e.g., peaky) currents from the motor current I.sub.MOTOR when the motor drive circuit **620** is drawing current directly from the batteries **660**, such that the battery current I.sub.BATT has a substantially DC magnitude. Although illustrated as the inductor **664**, in other examples the motor drive unit **600** may include a different filter circuit or the second filter circuit may be omitted. For instance, the motor drive unit **600** may also include a diode D**666** coupled between circuit common and the junction of the first switching circuit **662** and the inductor **664**. The diode D**666** may be configured to conduct current through the inductor **664** and the bus capacitor C.sub.BUS when the first switching circuit **662** is non-conductive and while the first switching circuit **662** is rendered conductive (e.g., gradually closed, for example, as described in more detail herein). Further, in some examples, the motor drive unit **600** may include an active filtering component, such as a filter circuit including a field-effect transistor (FET), that is configured to perform active and/or synchronous rectification. For instance, the motor drive unit **600** may include the active filtering component instead of the diode D**666**.

(82) In some examples, the control circuit **630** may render the first switching circuit **662** conductive (e.g., and render and/or maintain the second switching circuit **668** non-conductive) when the control circuit **630** has received an input or command to operate the motor **610** and has determined that the magnitude of the storage voltage V.sub.S of the energy storage element **654** (e.g., based on the magnitude of the scaled storage voltage V.sub.SS) is depleted below a threshold (e.g., does not have enough energy to complete a movement or an amount of movement of the covering material). For example, the control circuit may determine if the energy storage element **654** has enough energy to complete a movement or an amount of movement of the covering material by comparing a present storage capacity of the energy storage element **654** (e.g., the storage voltage V.sub.S) to a movement capacity threshold. For example, the movement capacity threshold may indicate a storage capacity sufficient to complete a full movement of the covering material from the fully-lowered position to the fully-raised position (e.g., a fixed threshold). In addition, the movement capacity threshold may be constant (e.g., such as 2.6 volts) or may vary, for example, depending on the amount of movement of the covering material required by the received command, such that the movement capacity threshold (e.g., a variable threshold) may indicate a storage capacity sufficient to complete the movement required by the received command.

(83) If the energy storage element **654** is not sufficiently charged (e.g., does not have enough energy to move the covering material), the control circuit may close the first switching circuit **662** (e.g., and render and/or maintain the second switching circuit **668** non-conductive) to allow the electrical load (e.g., the motor) to draw current directly from the batteries **660**. Closing the first switching circuit **662** (e.g., and rendering and/or maintaining the second switching circuit **668** non-conductive) may bypass the energy storage element **654**, such that the stored energy of the energy storage element **654** is not used for driving the motor **610** to move the covering material. As described herein, “closing” a switching circuit may refer to rendering the switching circuit conductive, while “opening” a switching circuit may refer to rendering the switching circuit non-conductive.

(84) As described in more detail herein, the control circuit **630** may operate in different modes, such as a first mode where the motor drive circuit **620** draws current (e.g., the motor current I.sub.MOTOR) from the batteries **660** (e.g., directly from the batteries via the inductor L**664**) to

control the power delivered to the motor **610** to control movement of the covering material, and operate in a second mode where the motor drive circuit **620** draws the current (e.g., the motor current I.sub.MOTOR) from the energy storage element **654** to control the power delivered to the motor **610** to control movement of the covering material. The control circuit **630** may render the first switching circuit **662** conductive and the second switching circuit **668** non-conductive to operate in the first mode, and render the first switching circuit **662** non-conductive and the second switching circuit **668** conductive to operate in the second mode.
(85) In some examples, the control circuit **630** may operate in the first mode until the battery voltage V.sub.BATT (e.g., the open-circuit battery voltage of the batteries) falls below a threshold voltage V.sub.TH (e.g., 1.4 V). After the battery voltage V.sub.BATT falls below the threshold voltage V.sub.TH, the control circuit **630** may render the first switching circuit **662** non-conductive and the second switching circuit **668** conductive to operate in the second mode to cause the motor drive circuit **620** to draw the current from the energy storage element **654** to control the power delivered to the motor **610** to control movement of the covering material. By operating in the second mode after the battery voltage V.sub.BATT falls below the threshold voltage V.sub.TH, the control circuit **630** may, for example, allow for more energy to be depleted out of the batteries **660** for use to control the motor **610** by drawing the current out of the batteries **660** at a level that is less than what is required to drive the motor **610**.
(86) Alternatively or additionally, the control circuit **630** may be configured to switch between the modes during operations of the motor **610** (e.g., during movements of the covering material between positions). For instance, the control circuit **630** may operate in the second mode when the motor current I.sub.MOTOR required by the motor drive circuit **620** to rotate the motor **610** is above a current threshold (e.g., 50-500 mA), and may operate in the first mode (e.g., switch to the first mode) when the motor current I.sub.MOTOR required by the motor drive circuit **620** to rotate the motor **610** is below the current threshold. An example of when this might occur is during the movement of the covering material between a fully-lowered position to a fully-raised position where the motor current I.sub.MOTOR needed to drive the motor **610** might be above the current threshold for at least the initial movement of the covering material, but when the covering material is close to the fully-raised position, the motor current I.sub.MOTOR may reduce below the current threshold and the control circuit **630** may operate in the first mode to drive the motor **610** from the batteries **660** (e.g., directly from the batteries **660** via the inductor L**664**). In some examples, the motor current I.sub.MOTOR may cross the current threshold during some, but not all, movements of the covering material.
(87) As noted above, in some examples, the motor drive unit **600** may be configured to operate in the first mode where the motor drive circuit **620** draws the battery current I.sub.BATT from the batteries **660** (e.g., directly from the batteries via the inductor L**664**) to control the power delivered to the motor **610** to control movement of the covering material, and/or operate in the second mode where the motor drive circuit **620** draws the current from the energy storage element **654** to control the power delivered to the motor **610** to control movement of the covering material. The motor drive unit **600** may be configured to switch between the first and second modes by controlling which (e.g., if either) of the first switching circuit **662** or the second switching circuit **668** is conductive (e.g., closed). At the time when the motor drive unit **600** enters the first mode where the motor drive circuit **620** draws the motor current I.sub.MOTOR from the batteries **660** (e.g., the first switching circuit **662** is being rendered conductive), the bus capacitor C.sub.BUS may not be charged to the magnitude of the battery voltage V.sub.BATT, which may result in a large pulse of current being drawn from the batteries **660** to charge the bus capacitor C.sub.BUS. This large pulse of current conducted through the batteries **660** may cause the magnitude of the battery voltage V.sub.BATT to dip, which in turn may causes the magnitude of the low-voltage supply voltage V.sub.CC to drop below a drop-out magnitude (e.g., 1.9V). In some instances, the magnitude of the battery voltage V.sub.BATT may dip in response to the large pulse of current conducted through the

batteries **660** due to the high equivalent series resistance (ESR) of the batteries (e.g., which may be alkaline batteries). This may cause the low-voltage circuitry of the motor drive unit **600** to become unpowered, become unfunctional, and/or reset, which may disrupt the operation of the motor drive unit **200** and/or movement of the covering material.

(88) As such, to avoid this situation, the control circuit **630** may be configured to gradually close the first switching circuit **662** when entering the first mode. For instances, the control circuit **630** may be configured to gradually change the first switching circuit **662** from a non-conductive state to a conductive state (e.g., over a time period of approximately 35-50 ms) to gradually close the first switching circuit **662**. For example, when the control circuit **630** is not controlling the movement of the covering material, the first and second switching circuits **662**, **668** may be open (e.g., non-conductive). The control circuit **630** may, for instance, receive an input or command to operate the motor **610**. In some examples, the control circuit **630** may determine (e.g., measure) the magnitude of the storage voltage V.sub.S (e.g., based on the scaled storage voltage V.sub.SS) and determine whether the magnitude of the storage voltage V.sub.S is greater than a threshold (e.g., the movement capacity threshold). When the magnitude of the storage voltage V.sub.S is greater than the movement capacity threshold (e.g., and prior to controlling the motor drive circuit **620** to generate the bus voltage V.sub.BUS at a boosted magnitude (e.g., approximately 5 volts) for powering the movement of the covering material), the control circuit **630** may close the second switching circuit **668** to charge the magnitude of the bus voltage V.sub.BUS to the magnitude of the storage voltage V.sub.S across the energy storage element **654** (e.g., approximately 3.5 volts). However, when the magnitude of the storage voltage V.sub.S is less than the movement capacity threshold, the control circuit **630** may gradually close the first switching circuit **662** to charge the magnitude of the bus voltage V.sub.BUS to the magnitude of the battery voltage V.sub.BATT. By gradually closing the first switching circuit **662**, the motor drive unit **600** may slowly charge up the bus capacitor C.sub.BUS and avoid any large peaks of current that may cause the aforementioned problems. After movement of the covering material is complete, the control circuit **630** may be configured to open at least one of the first switching circuit **662** and/or the second switching circuit **668** that was closed to power the motor **610**.

(89) In some instance, the control circuit **630** may be configured to pulse width modulate the first switch control signal V.sub.SW1 (e.g., to generate a pulse width modulated (PWM) gate signal at a gate of the first switching circuit **662**) to gradually close the first switching circuit **662** (e.g., using open-loop control). The control circuit **630** may be configured to generate the PWM gate signal (e.g., which may be generated as a PWM sequence that is applied to the first switch control signal V.sub.SW1) at a constant frequency but with an increasing on-time from one period to the next to gradually close the first switching circuit **662**. For example, the control circuit **630** may be configured to increase an on-time of a duty cycle of the PWM gate signal (e.g., the first switch control signal V.sub.SW1) from one cycle to the next to gradually close the first switching circuit **662**, until for example, the control circuit **630** renders the first switching circuit **662** continuously conductive. As such, the control circuit **630** may reduce the current conducted when charging of the bus capacitor C.sub.BUS as the first switching circuit **662** is being rendered conductive by pulse width modulating the first switch control signal V.sub.SW1. In some examples, the control circuit **630** may be configured to pulse width modulate the first switch control signal V.sub.SW1 at a frequency of approximately 25 kHz (e.g., a period of 40 μs) with an on-time that may vary between a minimum on-time (e.g., approximately 2.5 μs) to a maximum on-time (e.g., approximately 31.75 μs). In some examples, the control circuit **630** may be configured to increase the on-time from the minimum on-time to the maximum on-time, and maintain the on-time at the maximum on-time for a period of time before generating a constantly conductive first switch control signal V.sub.SW1 (e.g., rendering the first switching circuit **662** fully and constantly conductive). For example, the control circuit **630** may be configured to increase the on-time by an adjustment step each time that the control circuit **630** increases the on-time such that the on-time increases linearly from the

minimum on-time to the maximum on-time. Finally, in some examples, the control circuit **630** may be configured to increase the on-time is a non-linear manner. Further, in examples where the frequency and period of the PWM gate signal are constant, the duty cycle DC of the PWM gate signal may increase at a corresponding rate as the on-time increases. For example, the duty cycle DC of the PWM gate signal may increase through the PWM sequence (e.g., within a range of approximately 6% duty cycle to approximately 80% duty cycle, for instance, based on a frequency of approximately 25 kHz (e.g., a period of 40 μs) with an on-time that may vary between a minimum on-time (e.g., approximately 2.5 μs) to a maximum on time (e.g., approximately 31.75 μs)).

(90) In some examples, the control circuit **630** may be configured to gradually decrease the impedance of the first switching circuit **662** from a non-conductive impedance (e.g., a high-impedance state) to a conductive impedance (e.g., a low-impedance state) to gradually close the first switching circuit **662**. For example, the non-conductive impedance of the first switching circuit **662** may be very large and/or an open circuit, and the conductive impedance of the first switching circuit **662** may be the drain-to-source on-resistance R.sub.DS-ON of one of the FETs of first switching circuit **662** (e.g., very small). For instance, the control circuit **630** may be configured to control the average impedance of the first switching circuit **662** to decrease from the non-conductive impedance to the conductive impedance. In examples where the first switching circuit **662** includes one or more FETs, the impendence of the first switching circuit **662** may be the drain-to-source on-resistance R.sub.DS-ON when the first switching circuit **662** is conductive and an average impedance Z of the first switching circuit **663** (e.g., over one period of the PWM gate signal) may be based on the present duty cycle DC of the PWM gate signal (e.g., Z=DC.Math.R.sub.DS-ON).

(91) Finally, in some examples, the control circuit **630** may be configured to control the impedance of the first power source switching circuit **662**, where the first power source switching circuit **662** may be a variable resistance circuit, which may be controlled to gradually close the first power source switching circuit **662**. For example, the first power source switching circuit **662** may comprise one or more FETs controlled in the linear region to control the impedance of the first power switching circuit **662**. In other examples, the first switching circuit **662** may comprise may be a variable resistance circuit that includes multiple paths (e.g., resistive paths) configured to be coupled in series between the batteries **660** and the motor drive circuit **620**. For example, each path may comprise one or more resistors coupled in series with a switch (e.g., a switching circuit, such as one or more FETs). In addition, at least one of the paths may not comprise any resistors (e.g., the path may be a short circuit coupled in series with a switch). In some examples, the variable resistance circuit may include four paths with a first path providing a high resistance, a second path providing a medium resistance, a third path providing a low resistance, and a fourth path providing no resistance. The variable resistance circuit may be controlled to gradually close the first power source switching circuit **662** by controlling the switches coupled in series with each of the paths to first couple the first path between the batteries **660** and the motor drive circuit **620**, next couple the second path between the batteries **660** and the motor drive circuit **620**, then couple third path between the batteries **660** and the motor drive circuit **620**, and finally couple the fourth path between the batteries **660** and the motor drive circuit **620**. In some examples, the variable resistance circuit may comprise two paths with a first path having a resistor coupled in series with a respective switch and a second path having a short circuit coupled in series with a respective switch.

(92) Although primarily described with reference to gradually closing the first switching circuit **662**, the control circuit **630** may be configured to gradually close the second switching circuit **668** (e.g., in addition to or as an alternative to gradually closing the first switching circuit **662**) to, for example, gradually charge the bus capacitor C.sub.BUS and reduce any inrush current that might otherwise occur.

(93) FIG. **5** is a block diagram of an example filter circuit **470** and an example power converter circuit **452** for charging an energy storage element **454** from one or more batteries **460**. For example, the filter circuit **470** and the power converter circuit **452** may be configured for use in the motor drive unit **600** of FIG. **4**, such that the power converter circuit **452**, the filter circuit **470**, the energy storage element **454**, and the batteries **460** may be examples of the power converter circuit **652**, the first filter circuit **670**, the energy storage element **654**, and the batteries **660**, respectively. FIG. **6** is an example of waveforms that illustrate an operation of an energy storage element, power converter circuit, and filter, such as the power converter circuit **452** and the filter circuit **470**.
(94) The filter circuit **470** may comprise an inductor L.sub.1 and a capacitor C.sub.1. The filter circuit **470** may be a low-pass filter. The filter circuit **470** may receive a battery voltage V.sub.BATT from the batteries **460**. The power converter circuit **452** may draw a battery current I.sub.BATT from the batteries **460** through the filter circuit **470**. The filter circuit **470** may filter high and/or low frequency components of the battery current I.sub.BATT and/or the battery voltage V.sub.BATT. For instance, the filter circuit **470** may smooth primary current I.sub.1 conducted by a primary winding of a transformer T**420** to generate a low-ripple battery current I.sub.BATT (e.g., smooth out the average battery current I.sub.BATT, for example as shown in FIG. **6**). It should be appreciated that, in some examples, the filter circuit **470** may be omitted from the motor drive circuit (e.g., the motor drive circuit **600**).
(95) The power converter circuit **452** may include a converter control circuit **410**, a transformer T**420**, a switch S**412**, and scaling circuits **420**, **430**. As shown in FIG. **5**, the power converter circuit **452** may comprise, for example, a flyback converter circuit. The transformer T**420** may be characterized by a turns ratio of N:1. The input of the power converter **452** may be coupled to the batteries **460** through the filter circuit **470**. The output of the power converter circuit **452** may be coupled to the energy storage element **454**. The power converter circuit **452** may receive the battery voltage V.sub.BATT through the filter circuit **470**. The power converter circuit **452** may be configured to conduct the battery current I.sub.BATT from the batteries **460** to charge the energy storage element **654** and produce a storage voltage V.sub.S across the energy storage element **454**. For example, the energy storage element **454** may comprise one or more supercapacitors, rechargeable batteries, and/or other suitable energy storage devices. As illustrated in FIG. **5**, the converter control circuit **410** may be a dedicated control circuit for the power converter circuit **452**. The converter control circuit **410** may comprise, for example, a microprocessor, a programmable logic device (PLD), a microcontroller, an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or any suitable processing device or control circuit. In some examples, the functions of the converter control circuit **410** may be handled by another control circuit, such as a control circuit for the entire motor drive unit (e.g., the control circuit **630** of the motor drive unit **600** of FIG. **4**).
(96) The converter control circuit **410** may operate to reduce the magnitude (e.g., the average magnitude) of the battery current I.sub.BATT drawn from the batteries **460** while maintaining the operation of the power converter circuit **452** in discontinuous conduction mode. The converter control circuit **410** may determine a desired average current I.sub.AVE (e.g. I.sub.IN) to be drawn from the batteries **460** and a duty cycle for controlling the power converter circuit **452**. In examples the current limit of the power converter circuit may define an upper bound of the battery current I.sub.BATT. For example, the converter control circuit **410** may determine the desired average current I.sub.AVE drawn from the batteries **460**. Based on the desired average current I.sub.AVE and the goal of maintaining operation in discontinuous mode, the converter control circuit **410** may control a duty cycle of the power converter circuit **452** (e.g., by controlling an on-time t.sub.ON and an operating period t.sub.PERIOD of the power converter circuit).
(97) The converter control circuit **410** may be configured to determine the magnitude of an input voltage V.sub.1 of the power converter circuit **452** and the magnitude of an output voltage V.sub.2 of the power converter circuit **452** (e.g., the supply voltage V.sub.S). The converter control circuit

410 may receive a first scaled storage voltage V.sub.SS1 through the scaling circuit **420**, and may receive a second scaled voltage V.sub.SS2 through the scaling circuit **430** (e.g., which may be an example of the scaling circuit **656**). The converter control circuit **410** may be configured to determine the magnitude of the input voltage V.sub.1 in response to the magnitude of the first scaled storage voltage V.sub.SS1, and to determine the magnitude of the output voltage V.sub.2 (e.g., the storage voltage V.sub.S) in response to the magnitude of the second scaled storage voltage V.sub.SS2 (e.g., which may be an example of the motor drive unit scaled voltage V.sub.SS). The second scaled voltage V.sub.SS2 may be indicative of the battery voltage V.sub.BATT.

(98) The converter control circuit **410** may be configured to control (e.g., enable and disable) the power converter circuit **452**, using the switch S**412** (e.g., a semiconductor switch such as a FET), to control the magnitude of the battery current I.sub.BATT drawn from the batteries **460** (e.g., towards the average current I.sub.AVE). The converter control circuit **410** may render the switch S**412** conductive for an on-time t.sub.ON and non-conductive for an off-time t.sub.OFF, such that the power converter circuit **452** operates at an operating period t.sub.PERIOD (e.g., t.sub.PERIOD=t.sub.ON+t.sub.OFF). The converter control circuit **410** may generate a flyback control signal V.sub.FC (e.g., a drive voltage) for rendering the switch S**412** conductive and non-conductive. The converter control circuit **410** may determine the on-time t.sub.ON and the operating period t.sub.PERIOD for the flyback control signal V.sub.FC based on the desired average current I.sub.AVE to be drawn from the batteries **460** and to ensure that the power converter circuit **452** operates in discontinuous conduction mode. For example, the converter control circuit **410** may determine the on-time t.sub.ON and the operating period t.sub.PERIOD using on the received scaled storage voltages V.sub.SS1 and V.sub.SS2. The converter control circuit **410** may render the switch S**412** conductive and non-conductive based in whole or in part on the duty cycle, the on-time t.sub.ON, the operating period t.sub.PERIOD, the received scaled storage voltages V.sub.SS1 and V.sub.SS2, the desired average current I.sub.AVE, or any combination thereof.

(99) The converter control circuit **410** may render the switch S**412** conductive by controlling the flyback control signal V.sub.FC to be high for the on-time t.sub.ON. During the on-time t.sub.ON, a primary winding of the transformer T**420** may be configured to conduct a primary current I.sub.1 thus charging a magnetizing inductance L.sub.2 of the transformer T**420**. For example, while the switch S**412** is conductive during the on-time t.sub.ON, the magnitude of the primary current I.sub.1 may rise with respect to time (e.g., linearly) until, for example, the end of the on-time t.sub.ON. The magnitude of the primary current I.sub.1 may reach a peak magnitude I.sub.1_pk at the end of the on-time t.sub.ON. At the end of the on-time t.sub.ON, the converter control circuit may drive the flyback control signal V.sub.FC low thereby rendering the switch **412** non-conductive. A secondary winding of the transformer T**420** may conduct a secondary current I.sub.2 (e.g., a charging current) through a diode D**440** to charge the energy storage element **454** during a first portion of the off-time, t.sub.OFF1. The secondary current I.sub.2 may begin at a peak magnitude and decrease with respect to time (e.g., linearly) until the magnitude reaches zero amps at the conclusion of the first portion of the off-time, t.sub.OFF1. At the end of the first portion of the off-time, t.sub.OFF1, the control circuit may maintain the switch S**412** non-conductive for a second portion of the off-time, t.sub.OFF2 (e.g., a dead time). The control circuit **410** may provide (e.g., determine) the second portion of the off-time, t.sub.OFF2 to maintain the operation of the power converter **452** in discontinuous conduction mode. At the end of the second portion of the off-time, t.sub.OFF2, the control circuit **410** may start another operating period and drive the flyback control signal V.sub.FC high to render the switch S**412** conductive for the on-time t.sub.ON of the next operating period t.sub.PERIOD. The control circuit **410** may drive the flyback control signal V.sub.FC low to render the switch **2412** non-conductive for the duration of the off-time t.sub.OFF1+t.sub.OFF2. Accordingly, the converter control circuit **410** may set the on-time t.sub.ON and the operating period t.sub.PERIOD (e.g., to set the second portion of the off-time,

t.sub.OFF2) to ensure that the power converter **452** operates in a manner to ensure the desired low average current I.sub.AVE (e.g., approximately 15 milliamps) may be drawn from the batteries **460** to extend the lifetime of the batteries **460**, while also maintaining operation of the power converter **452** in discontinuous conduction mode.

(100) It should be appreciated that the magnitude of the input voltage V.sub.1 may vary with time (e.g., over the lifetime of the batteries **460** or upon installation of new batteries). Further, the magnitude of the storage voltage V.sub.S may vary with time, for example, based on the operation of the power converter circuit **452** (e.g., the manner in which the switch S**412** is driven), or as the energy storage element **454** charges and discharges. The lengths of the on-time t.sub.ON and the operating period t.sub.PERIOD may be determined based on changes to the magnitude of the input voltage V.sub.1 and the output voltage V.sub.2 (e.g., and may vary because of the changes to the magnitude of the input voltage V.sub.1 and the output voltage V.sub.2). The control circuit may set the length of the operating period t.sub.PERIOD based on the on-time t.sub.ON and the first portion of the off-time t.sub.OFF using a scaling factor α, e.g.,

t.sub.PERIOD=α.Math.(t.sub.ON+t.sub.OFF1)=(t.sub.ON+t.sub.OFF1+t.sub.OFF2). (Equation 1)

For example, the scaling factor a may be a constant value that is set is to be greater than one, such that the magnitude of the secondary current I.sub.2 may always reach zero amps prior to the end of the operating period t.sub.PERIOD and the power converter circuit **452** may operate in the discontinuous conduction mode. The operating period may be regarded as the duration of the combination of the on-time t.sub.ON and the off-time (e.g., t.sub.OFF=t.sub.OFF1+t.sub.OFF2). The on-time t.sub.ON may be determined such that it is proportional to a ratio where the inductance L.sub.2 is multiplied by the desired average current I.sub.AVE divided by the current voltage V.sub.1. The on-time t.sub.ON may be determined (e.g., calculated) based on the magnitude of the input voltage V.sub.1, the magnitude of the output voltage V.sub.2, the desired average current I.sub.AVE, the magnetizing inductance L.sub.2 of the transformer T**420** and/or the scaling factor α, e.g.,

t.sub.ON=(2.Math.I.sub.AVE.Math.L.sub.2α.Math.[N.Math.V.sub.2+V.sub.1])/(N.Math.V.sub.1.Math.V.sub.2). (Equation 2)

The first portion of off-time t.sub.OFF1 may vary with time based on the peak magnitude I.sub.1_pk, the magnetizing inductance L.sub.2 of the transformer T**420**, the turns ratio N:1 of the transformer T**420**, and the magnitude of the output voltage V.sub.2. For example, the operating period t.sub.PERIOD may be determined (e.g., calculated) based on the magnitude of the input voltage V.sub.1, the magnitude of the output voltage V.sub.2, the desired average current I.sub.AVE, and/or the scaling factor α, e.g.,

t.sub.PERIOD=(2.Math.I.sub.AVE.Math.L.sub.2α.sup.2[N.Math.V.sub.2+V.sub.1].sup.2)/(V.sub.1.Math.[N.Math.V.sub.2].sup.2). (Equation 3)

The control circuit **410** may be configured to periodically adjust the on-time t.sub.ON and the operating period t.sub.PERIOD at an interval that is greater than the operating period t.sub.PERIOD of the power converter circuit. For example, the interval may be approximately one second when the energy storage element **454** comprises one or more supercapacitors, and approximately one minute when the energy storage element **454** comprises one or more rechargeable batteries.

(101) Further, as described herein, a control circuit of a control device (the converter control circuit **410** and/or the control circuit **630**) may use a lockout flag to indicate that the storage level of the energy storage element (e.g., as indicated by the storage voltage V.sub.S) is depleted to a level at which the energy storage element may not be capable of driving the motor. For example, the lockout flag may indicate that the storage level of the energy storage element is and/or was less than a threshold level (e.g., if the magnitude of the storage voltage V.sub.S is and/or was less than a

movement charge threshold, such as 2.6 volts) and has not exceeded a high threshold TH.sub.HI (e.g., approximately 3.6 volts). As described herein, the control circuit may set the lockout flag when the control circuit receives a command to move and determines that there is not enough charge in the energy storage element (e.g., the storage voltage V.sub.S is less than the movement charge threshold, e.g., 2.6 volts). The control circuit may set the lockout flag in response to entering a locked charging session. As also described herein, the locked charging session may prevent the control circuit from driving the motor from the energy storage element when the magnitude of the storage voltage of the energy storage element is between the movement capacity threshold and the high threshold TH.sub.HI (e.g., when the magnitude of the storage voltage of the energy storage element is increasing from the movement capacity threshold to the high threshold TH.sub.HI). The locked charging session may ensure that the energy storage element is fully charged (e.g., the storage voltage V.sub.S is greater than the high threshold TH.sub.HI) before the energy storage element is used to drive the motor.
(102) FIG. **7** is a flowchart of an example procedure **700** for charging an energy storage element (e.g., the energy storage element **454**, **654**). The procedure **700** may be executed by a control circuit of a control device, for example, the converter control circuit **410** and/or the control circuit **630**. The procedure **700** may be used to enable or disable a power converter circuit (e.g., the power converter circuit **452**, **652**), such that the power converter circuit may selectively charge the energy storage element from one or more batteries (e.g., the batteries **460**, **660**). The procedure **700** may ensure that the energy storage element begins charging (e.g., the power converter circuit begins drawing current from the batteries) when the energy storage element has discharged to a level where it may not have sufficient charge. The procedure **700** may ensure that the power converter circuit discontinues charging once the energy storage element has been charged to a sufficient level. The charging may occur at a rate such that the energy storage element draws a desired average current over time from the batteries, where the desired average current is below a threshold and/or is sufficiently low (e.g., approximately 15 milliamps) that the battery or batteries the energy storage element draws energy from may have an extended lifetime.
(103) The control circuit may execute (e.g., periodically execute) the control procedure **700** at **710**. At **712**, the control circuit may determine whether a charging flag is set. As described herein, the charging flag may indicate whether a charging session is active and the power converter circuit is enabled (e.g., if the power converter circuit is presently charging the energy storage element). If the charging flag is not set at **712**, the control circuit may determine if the magnitude of the storage voltage V.sub.S is less than a low-side threshold TH.sub.LO (e.g., approximately 3.1 volts) at **714**. In some cases, the magnitude of the storage voltage V.sub.S may drop below the low-side threshold TH.sub.LO when the energy storage element is depleted after powering a movement or movements of the covering material. If the magnitude of the storage voltage V.sub.S is not less than the low-side threshold TH.sub.LO, the control circuit may exit the procedure **700**. If the magnitude of the storage voltage V.sub.S is less than the low-side threshold TH.sub.LO, the control circuit may start a charging session by enabling the power converter circuit at **716**. Enabling the power converter circuit may commence charging of the energy storage element (e.g., to ensure the energy storage element possesses sufficient charge for a movement or movements of the covering material).
(104) At **718**, the control circuit may set the charging flag. As noted herein, the charging flag may indicate when a charging session is active. The control circuit may set the charging flag in response to starting the charging session (e.g., when the magnitude of the storage voltage V.sub.S is less drops below the low-side threshold TH.sub.LO) and enabling the power converter circuit. As described herein, in some examples, the control circuit may disable the power converter circuit in response to the reception of a command to move (e.g., as described with reference to FIG. **9**) based on the status of the charging flag (e.g., when the charging flag is set during the charging session). For example, the control circuit may ensure that the batteries are not charging the energy storage element (e.g., via the power converter circuit) while also providing current to the motor drive

circuit **610** to control movement of the covering material.

(105) In some examples, if, while the charging flag is set (e.g., the power converter circuit is enabled) and the control circuit is charging the energy storage element, the control circuit receives an input or command to control power delivered to a load (e.g., operate a motor of the control device), the control circuit may determine whether the storage voltage V.sub.S of the energy storage element is above the movement capacity threshold. If the storage voltage V.sub.S is above the movement capacity threshold, the control circuit may operate in a second mode and cause the drive unit to draw the current from the energy storage element to control the power delivered to the load. However, if the storage voltage V.sub.S is not above the movement capacity threshold, the control circuit may operate in the second mode and cause the drive unit to draw the current from the batteries to control the power delivered to the load. Further, and for example, the control circuit may not utilize the locked charging session and/or the lockout flag, in some instances.

(106) If the charging flag is set at **712** (e.g., the power converter circuit is enabled), the control circuit may determine if the magnitude of the storage voltage V.sub.S is greater than or equal to a high threshold TH.sub.HI (e.g., approximately 3.6 volts) at **720**. The magnitude of the storage voltage V.sub.S may be greater than or equal to the high-side threshold TH.sub.HI if the energy storage element **654** is charged (e.g., to a sufficient level). If the control circuit determines that the magnitude of the storage voltage V.sub.S is not greater than or equal to the high-side threshold TH.sub.HI at **720**, the control circuit may exit the control procedure **700**. If the control circuit determines that the magnitude of the storage voltage V.sub.S is greater than or equal to the high-side threshold TH.sub.HI at **720**, the control circuit may end the charging session by disabling the power converter circuit at **722**. Disabling the power converter circuit may cause the energy storage element to cease charging (e.g., if the energy storage element possesses sufficient charge for a movement or movements of the covering material).

(107) At **724**, the control circuit may clear the charging flag. As noted herein, the charging flag may indicate when a charging session is active. At **726**, the control circuit may determine whether the lockout flag is set. The lockout flag may indicate that the storage level of the energy storage element (e.g., as indicated by the storage voltage V.sub.S) is depleted a level at which the energy storage element may not be capable of driving the motor. For example, the lockout flag may indicate that the storage level of the energy storage element is and/or was less than a threshold level (e.g., if the magnitude of the storage voltage V.sub.S is and/or was less than a movement charge threshold, such as 2.6 volts) and has not exceeded the high threshold TH.sub.HI. As described herein, the control circuit may set the lockout flag when the control circuit receives a command to move and determines that there is not enough charge in the energy storage element (e.g., the storage voltage V.sub.S is less than the movement charge threshold, e.g., 2.6 volts). The control circuit may set the lockout flag in response to entering a locked charging session. For instance, when the storage level of the energy storage element (e.g., as indicated by the storage voltage V.sub.S) is less than the threshold level (e.g., when the magnitude of the storage voltage V.sub.S is less than the movement charge threshold), the control circuit may enable the power converter circuit to commence charging of the energy storage element from the batteries when the control circuit is not driving the motor. While charging the energy storage element, the control circuit may ensure that the motor drive circuit does not draw current from the energy storage element to control the power delivered to the motor. As such, the locked charging session may prevent the control circuit from driving the motor from the energy storage element when the magnitude of the storage voltage of the energy storage element is between the movement capacity threshold and the high threshold TH.sub.HI (e.g., when the magnitude of the storage voltage of the energy storage element is increasing from the movement capacity threshold to the high threshold TH.sub.HI). The locked charging session may ensure that the energy storage element is fully charged (e.g., the storage voltage V.sub.S is greater than the high threshold TH.sub.HI) before the energy storage element is used to drive the motor.

(108) If the control circuit determines that the lockout flag is set at **726**, the control circuit may clear the lockout flag at **728** (e.g., to end the locked charging session), and exit the procedure **700**. However, if the control circuit determines that the lockout flag is not set at **726**, the control circuit may exit the procedure **700**. As such, the control circuit may ensure that the lockout flag cleared after the control circuit determines that the storage voltage of the energy storage element is above the high threshold TH.sub.HI. Finally, it should be appreciated that in some examples, the use of the charging flag (e.g., **718** and **74**) and/or the lockout flag (e.g., **726** and **728**) may be omitted from the procedure **700**.
(109) FIG. **8** is a flowchart of an example procedure **800** for charging an energy storage element (e.g., the energy storage element **454**, **654**) from one or more batteries (e.g., the batteries **460**, **660**). The procedure **800** may be executed by a control circuit of a control device, for example, the converter control circuit **410** and/or the control circuit **630**. The procedure **800** may be used to ensure a power converter circuit (e.g., the power converter circuit **452**, **652**) draws current from the batteries to charge the energy storage element such that a battery current drawn from the batteries does is at a low level (e.g., the desired average current I.sub.AVE). The control circuit may execute the procedure **800** periodically. For example, the control circuit may execute the procedure at an interval (e.g., an execution period) that is greater than the operating period of the power converter circuit. When the energy storage element comprises one or more supercapacitors, the control circuit may execute the procedure **800**, for example, every one second. When the energy storage element comprises one or more rechargeable batteries, the control circuit may execute the procedure **800**, for example, every one minute.
(110) The control circuit may start the control procedure **800** at **810**. At **812**, the control circuit may determine a desired average current (e.g., the desired average current I.sub.AVE) to be drawn from the batteries **660**. For example, the desired average current may be stored in memory in the control device, and the control circuit may retrieve the desired average current from the memory at **812**. At **814**, the control circuit may determine present magnitudes for an input voltage V.sub.1 of the power converter circuit and an output voltage V.sub.2 of the power converter circuit (e.g., which may be a storage voltage V.sub.S across the energy storage element). For example, the control circuit may sample the magnitude of the first scaled storage voltage V.sub.SS1 to determine the magnitude of the input voltage V.sub.1 and sample the magnitude of the second scaled storage voltage V.sub.SS2 to determine the magnitude of the output voltage V.sub.2 at **814**. At **816**, the control circuit may determine an on-time t.sub.ONand an operating period t.sub.PERIOD to use to control the power converter circuit. For example, the control circuit may determine the on-time t.sub.ON and the operating period t.sub.PERIOD at **816** based on the magnitude of the input voltage V.sub.1 and the magnitude of the output voltage V.sub.2 (e.g., as determined at **814**). For example, the control circuit may calculate the on-time t.sub.ON and the operating period t.sub.PERIOD using equations 2 and 3 as shown above. At **818**, the control circuit may generate a drive voltage (e.g., the flyback control signal V.sub.FC) to render a semiconductor switch of the power converter circuit (e.g., the switch S**412**) conductive for the duration of the determined t.sub.ON and non-conductive for the duration of a time period (e.g., an off-time) that is equal to the determined operating period t.sub.PERIOD minus the determined on-time t.sub.ON. Controlling the semiconductor switch of the power converter circuit to be conductive for the on-time t.sub.ON and non-conductive for the off-time t.sub.OFF (e.g., t.sub.OFF=t.sub.OFF1+t.sub.OFF2) may enable the power converter circuit to draw the desired average current from the DC power source (e.g., the batteries **660**) while the power converter circuit operates in discontinuous conduction mode.
(111) FIG. **9** is a flowchart of an example procedure **900** for selectively powering an electrical load from one or more batteries (e.g., the batteries **460**, **660**) or an energy storage device (e.g., the energy storage device **454**, **654**). The procedure **900** may be executed by a control circuit of a control device, for example, the converter control circuit **410** and/or the control circuit **630**. The

energy storage device may be configured to charge from the one or more batteries of the control device. The control circuit may be configured to control a first switch (e.g., a bypass switch, such as the first switching circuit **662**) or a second switch (e.g., an energy storage element switch, such as the second switching circuit **668**) for selectively powering the electrical load from the one or more batteries or the energy storage device. The control circuit may control the first switch and the second switch to either power the electrical load from the energy storage element or directly from the batteries (e.g., via a filter circuit, such as the inductor L**664**).
(112) In some examples, the procedure **900** may be used to prevent an inrush current from the batteries to a bus capacitor of the control device (e.g., the bus capacitor C.sub.BUS), as described herein, which may cause, for example, the magnitude of the low-voltage supply voltage V.sub.CC to drop below a drop-out magnitude (e.g., 1.9V), and the low-voltage circuitry of the motor drive unit **600** to become unpowered, become unfunctional, and/or reset, which may disrupt the operation of the control device. Alternatively or additionally, the procedure **900** may be used to ensure that the energy storage element has sufficient energy (e.g., 100 Joules) for powering the electrical load (e.g., for driving a motor to move a covering material of a motorized window treatment). Ensuring the energy storage element has sufficient energy to move the covering material prior to starting to move the covering material may prevent an interruption (e.g., a bump) or slowing in the movement of the covering material (e.g., if the control circuit has to close the bypass switch to change from powering the electrical load from the energy storage element to powering the electrical load from the batteries).
(113) The control circuit may start the control procedure **900** at **910**, e.g., in response to receiving an input or command (e.g., via the communication circuit **642** and/or the user interface **644**) to operate a motor of the control device (e.g., the motor **610**). At **912**, the control circuit may determine if the received command is a command to move a covering material. If the received command is not a command to move the covering material, the control circuit may exit the procedure **900**. Further, it should be appreciated that in some examples, prior to receiving the command to move, the first and second switches may both be in the open position (e.g., non-conductive).
(114) If the received command is a command to move the covering material at **912**, the control circuit may determine whether a charging flag is set at **914**. As described herein, the charging flag may indicate whether a charging session is active and the power converter circuit (e.g., the power converter circuit **652**) is enabled when a command to operate the motor is received (e.g., whether the batteries are presently charging the energy storage element when the control procedure **900** is executed). In some examples, the control circuit may set the charging flag in response to enabling the power converter circuit when the magnitude of the storage voltage V.sub.S drops below the low-side threshold TH.sub.LO (e.g., as described with reference to FIG. **7**). As such, the charging flag may indicate whether a charging session is active. If the control circuit determines that the charging flag is set at **914**, the control circuit may disable the power converter circuit at **916** (e.g., prior to controlling the motor in response to the received command). Otherwise, the control circuit may proceed directly to **918**.
(115) At **918**, the control circuit may determine whether the lockout flag is set. The lockout flag may indicate that indicate that a locked charging session is active. For example, the lockout flag may be set when the energy storage element is or was being charged by the batteries, and that the storage capacity of the energy storage element (e.g., as indicated by the storage voltage V.sub.S) is less than the high threshold TH.sub.HI (e.g., as described with reference to FIG. **7**). For example, the lockout flag may indicate that the storage level of the energy storage element (e.g., the magnitude of the storage voltage V.sub.S) is or was less than a threshold level (e.g., the movement capacity threshold, such as 2.6 volts) when the control circuit received a previous command to control the motor, and in turn, that the energy storage element was being charged by the batteries but that the storage voltage V.sub.S did not charge to a level that exceeded the high threshold

TH.sub.HI.
(116) If the control circuit determines that the lockout flag is not set at **918**, the control circuit may determine if the energy storage element is sufficiently charged (e.g., has enough energy to complete a movement or an amount of movement of the covering material) at **920**. For example, the control circuit may determine if the energy storage element has enough energy to complete a movement or an amount of movement of the covering material at **914** by comparing a present storage capacity of the energy storage element (e.g., as indicated by the storage voltage V.sub.S) to a threshold (e.g., by comparing the magnitude of the storage voltage V.sub.S to a movement charge threshold, such as 2.6 volts). The movement charge threshold may indicate a storage level sufficient to complete a full movement of the covering material from the fully-lowered position to the fully-raised position (e.g., a fixed threshold). In addition, in some examples, the movement charge threshold may vary depending on the amount of movement of the covering material required by the received command, such that the movement capacity threshold (e.g., a variable threshold) may indicate a storage level sufficient to complete the movement required by the received command.
(117) If the control circuit determines that the energy storage element is sufficiently charged at **920**, the control circuit may close the second switch at **922**. For example, if the magnitude of the storage voltage V.sub.S (e.g., based on the scaled storage voltage V.sub.SS) is greater than the movement capacity threshold (e.g., the energy storage element is sufficiently charged), the control circuit may close the second switch (e.g., the second switching circuit **668**) to charge the magnitude of a bus voltage of the bus capacitor to the magnitude of the storage voltage V.sub.S across the energy storage element (e.g., approximately 3.5 volts) at **922**.
(118) If the control circuit determines that the energy storage element is not sufficiently charged at **920**, the control circuit may set the lockout flag at **924** (e.g., to indicate that a locked charging session is active). For example, if the magnitude of the storage voltage V.sub.S (e.g., based on the scaled storage voltage V.sub.SS) is less than the movement charge threshold (e.g., the energy storage element is not sufficiently charged), the control circuit may set the lockout flag at **924**. If the control circuit determines that the lockout flag is set at **918**, the control circuit may proceed to **926**. For instance, by proceeding to **926** from **918**, the control circuit may ensure that the motor drive circuit does not draw current from the energy storage element to control the power delivered to the motor in response to receiving a command to move (e.g., at **912**), for example, even if the magnitude of the storage voltage V.sub.S is greater than the movement charge threshold. As such, the locked charging session may prevent the control circuit from driving the motor from the energy storage element when the storage voltage of the energy storage element is between the movement capacity threshold and the high threshold TH.sub.HI (e.g., and the energy storage element is or was being charged by the batteries). The locked charging session may ensure that the energy storage element is fully charged (e.g., the magnitude of the storage voltage V.sub.S has risen above the high-side threshold TH.sub.HI) before the energy storage element is used to drive the motor.
(119) After setting the lockout flag at **924** or in response to determining that the lockout flag is set at **918**, the control circuit may gradually close the first switch (e.g., the first switching circuit **662**) to charge the magnitude of a bus voltage of the bus capacitor to the magnitude of the battery voltage V.sub.BATT at **926**. The control circuit may gradually decrease the impedance of the first switch from a non-conductive impedance to a conductive impedance to gradually close the first switch. Further, in some examples, the control circuit may generate a PWM gate signal to gradually close the first switch (e.g., using open-loop control). For instance, the control circuit may pulse width modulate a first switch control signal (e.g., the first switch control signal V.sub.SW1) that is used to render the first switch (e.g., the first switching circuit **662**) conductive. The control circuit may be configured to pulse width modulate the first switch control signal at a constant frequency but with an increasing on-time to gradually close the first switch. By gradually closing the first switch, the control circuit may slowly charge up the bus capacitor and avoid any large peaks of current that, for example, may cause the aforementioned problems.

(120) In some examples, the procedure may not include the lockout flag and/or locked charging session, and in turn the procedure may not include **918** and **924**. In such instances, the control circuit may not enter into the locked charging session, and instead, the control circuit may compare the magnitude of the storage voltage V.sub.S to the movement capacity threshold (e.g., each time that a command is received) to determine whether the close the first switch or the second switch. Accordingly, the control circuit may close the second switch at **922** when the energy storage element is sufficiently charged (e.g., the magnitude of the storage voltage V.sub.S is greater than the movement charge threshold) or may close the first switch at **926** when the energy storage element is not sufficiently charged (e.g., the magnitude of the storage voltage V.sub.S is less than the movement charge threshold). And in these examples, the control circuit may not use a locked charging session.
(121) After closing the second switch at **922**, the control circuit may rotate the motor to move the covering material at **928** (e.g., by drawing current from the energy storage element and not drawing any current directly from the batteries). After closing the first switch at **926**, the control circuit may rotate the motor to move the covering material at **918** (e.g., by drawing current from the batteries and not the energy storage element). For example, closing the first switch may allow the electrical load (e.g., the motor) to draw current from the batteries. Closing the first switch may bypass the energy storage element, such that the stored energy of the energy storage element is not required for moving the covering material.
(122) The control circuit may continue rotating the motor at **928** until the movement of the covering material is complete at **930**. When the movement of the covering material is complete at **930**, the control circuit may open the first switch at **932** (e.g., assuming the first switch was closed) and open the second switch at **934** (e.g., assuming the second switch was closed). At **936**, the control circuit may determine whether the charging flag is set. If the charging flag is not set, the control circuit may exit the procedure **900**.
(123) If the control circuit determines that the charging flag is set at **936**, the control circuit may wait a delay period at **938** before enabling the power converter circuit at **940**, and the control circuit may exit the procedure **900**. When the power converter circuit is enabled at **940**, the power converter circuit may be configured to charge the energy storage element from the batteries. The delay period may be, for example, a predetermined time period (e.g., 5 minutes, 10 minutes, etc.). Further, in some examples, the delay period may be a first delay period (e.g., approximately 10 minutes) when the motor current was drawn (e.g., the motor was driven) from the energy storage element and not from the batteries (e.g., when the second switch is closed at **922**), or a second delay period (e.g., approximately 5 minutes) when the motor current was drawn (e.g., the motor was driven) from the batteries and not from the energy storage element (e.g., when the first switch is closed at **926**). The delay period may be used, for example, to allow the battery voltage and/or storage voltage of the energy storage element to stabilize after a movement of the covering material so that the control circuit can determine an accurate voltage measurement, for example, before starting and/or resuming to charge the energy storage element from the batteries.
(124) FIG. **10** is a flowchart of an example procedure **1000** for driving an electrical load (e.g., a motor that controls movements of a covering material of a motorized window treatment) by drawing current from one or more batteries (e.g., the batteries **460**, **660**) or from an energy storage element (e.g., the energy storage element **454**, **654**). The procedure **1000** may be executed by a control circuit of a device, for example, the converter control circuit **410** and/or the control circuit **630** of the motor drive unit **600**. The procedure **1000** may be used to change into a first mode or a second mode. The control circuit may execute the procedure **1000** periodically as a maintenance procedure. Alternatively or additionally, the control circuit may execute the procedure **1000** in response to a command to drive the electrical load (e.g., power the motor to move a covering material of the motorized window treatment).
(125) The control circuit may start the control procedure **1000** at **1010**. At **1012**, the control circuit

may compare the magnitude of a battery voltage of the one or more batteries (e.g., the battery voltage V.sub.BATT) to a threshold voltage V.sub.TH (e.g., 1.4 V). If the control circuit determines that the battery voltage V.sub.BATT (e.g., the open-circuit battery voltage of the batteries) is greater than the threshold voltage at **1012**, the control circuit may disable a power converter (e.g., the power converter **652**) at **1014**. The device may include a switch (e.g., a bypass switch, such as the first switching circuit **662**), and in such instances, the control circuit may ensure the switch is closed at **1016**. If the switch is open, the control circuit may generate a switch control signal (e.g., the first switch control signal V.sub.SW1) for rendering the switch conductive at **1016**.
(126) At **1018**, the control circuit may operate in a first mode where, upon receiving a command to drive the electrical load, the control circuit may draw current (e.g., the motor current I.sub.MOTOR) from the batteries (e.g., directly from the batteries via a filter circuit, such as the inductor L**664**) to power the electrical load. Accordingly, the control circuit may disable the power converter at **1014**, and close the switch at **1016** to bypass an energy storage element (e.g., the energy storage element **654**) of the device and to allow the drive unit to draw current directly from the batteries when the magnitude of the battery voltage is greater than the threshold voltage V.sub.TH. The procedure **1000** may then exit.
(127) If the control circuit determines that the battery voltage V.sub.BATT (e.g., the open-circuit battery voltage of the batteries) is less than the threshold voltage V.sub.TH at **1012**, the control circuit may enable the power converter at **1020**. In some examples, the control circuit may enable the power converter to charge the energy storage element. However, in some examples, if the energy storage element is already charged, then **1020** may be omitted. At **1022**, the control circuit may render the switch (e.g., the first switching circuit **662**) non-conductive. Further, in some examples, the device may include a second switch (e.g., such as the second switching circuit **668**), and the control circuit may ensure that the second switch is closed at **1022**.
(128) At **1024**, the control circuit may operate in a second mode to cause the drive unit to draw the current from the energy storage element to control the power delivered to the electrical load (e.g., instead of the batteries) at **1024**, and the procedure **1000** may exit. Accordingly, the control circuit may enable the power converter at **1020**, and open the switch at **1022** to allow the drive unit to draw current directly from the energy storage element to control the power delivered to the electrical load when the magnitude of the battery voltage is less than the threshold voltage V.sub.TH. Further, in some examples, when operating in the second mode, the control circuit may be configured to conduct the battery current V.sub.BATT from the batteries to charge the energy storage element (e.g., as described herein), but the control circuit may drive the load using the energy stored within the energy storage element.
(129) By operating in the second mode after the battery voltage V.sub.BATT falls below the threshold voltage V.sub.TH, the control circuit may, for example, allow for more energy to be depleted out of the batteries for use to control the load by drawing the current out of the batteries at a voltage level that is less than what is required to control the load (e.g., and also leverage the higher voltage potential (e.g., reduced voltage drop) provided by the energy storage element). Further, in some examples, the battery voltage V.sub.BATT may be greater than the threshold voltage V.sub.TH when the batteries are relatively new, and after many uses of directly powering a peaky load, the battery voltage V.sub.BATT may fall below the threshold voltage V.sub.TH. So by operating the procedure **1000**, the control circuit may use the batteries when they have a larger open-circuit battery voltage, and switch to use of the energy storage element when the open-circuit battery voltage falls below the threshold voltage V.sub.TH.
(130) Alternatively or additionally, the control circuit may be configured to switch between modes during operations of the motor. For instance, the control circuit may be configured to operate in the first mode of operation when the current needed by the motor to control movement of the covering material is below a threshold current, and operate in the second mode of operation when a current needed by the motor to control movement of the covering material is above the threshold current.

(131) FIG. **11** is a flowchart of an example procedure **1100** for driving an electrical load (e.g., a motor that controls movements of a covering material of a motorized window treatment) by drawing current from one or more batteries (e.g., the batteries **460**, **660**) or from an energy storage element (e.g., the energy storage element **454**, **654**). The procedure **1100** may be executed by a control circuit of a device, for example, the converter control circuit **410** and/or the control circuit **630** of a motorized window treatment. The procedure **1100** may be used to switch between a first mode and a second mode. The control circuit may execute the procedure **1100** periodically. Alternatively or additionally, the control circuit may execute the procedure **1100** in response to a command to drive the electrical load (e.g., power the motor to move a covering material of the motorized window treatment). If performed in response to a command to drive the electrical load, then **1112** may be omitted.

(132) The control circuit may start the control procedure **1100** at **1110**. When starting the control procedure **1100**, a switch (e.g., a bypass switch, such as the first switching circuit **662**) may be in the open position (e.g., and the second switching circuit **668** may be in the closed position). At **1112**, the control circuit may determine whether it received a command to drive the electrical load (e.g., a command to drive a motor to move a covering material of a motorized window treatment). If the control circuit determines that it did not receive a command to drive the electrical load at **1112**, then the control procedure **1100** may exit. If the control circuit determines that it received a command to drive the electrical load at **1112**, then the control circuit may determine whether a drive current required to drive the electrical load (e.g., the motor current I.sub.MOTOR required by the motor drive circuit **620** to rotate the motor **610**) is less than a current threshold I.sub.TH (e.g., approximately 50-500 mA).

(133) If the control circuit determines that the motor current I.sub.MOTOR is less than the current threshold I.sub.TH at **1114**, the control circuit may close the switch to operate in the first mode at **1116**. In some examples, the switch (e.g., the first switching circuit **662**) may already be closed, and in such instances, **1116** may be omitted. In some instances, the control circuit may be required to open a second switch (e.g., the second switching circuit **668**). At **1118**, the control circuit may draw the motor current I.sub.MOTOR from the batteries (e.g., directly from the batteries via a filter circuit, such as the inductor L**664**) to drive the electrical load. At **1126**, the control circuit may determine whether the driving of the electrical load is completed (e.g., whether a movement of a covering material as directed by the received command is complete). If so, the procedure **1100** may exit. However, if the driving of the load is not complete at **1126**, then the procedure **1100** may return to **1114**.

(134) If the control circuit determines that the motor current I.sub.MOTOR is greater than the current threshold I.sub.TH at **1114**, the control circuit may control a power converter (e.g., the power converter **652**) at **1120**, for example, to charge the energy storage element. However, in some examples, if the energy storage element is already charged, then **1120** may be omitted. At **1122**, the control circuit may render the switch (e.g., the first switching circuit **662**) non-conductive (e.g., and render the second switching circuit **668** conductive) to operate in the second mode. At **1224**, the control circuit may cause the device to draw the current from the energy storage element to control the power delivered to the electrical load (e.g., instead of the batteries). When operating in the second mode, the control circuit may be configured to conduct the battery current V.sub.BATT from the batteries to charge the energy storage element, for example, as described herein, but the control circuit may drive the load using the energy stored within the energy storage element. The control circuit may determine whether the driving of the load is completed at **1126**. If so, the procedure **1100** may exit.

(135) Accordingly, through the procedure **1100**, the control circuit may operate in the second mode when the magnitude of the motor current I.sub.MOTOR required by the motor drive circuit to rotate the motor is above the current threshold I.sub.TH (e.g., approximately 50-500 mA), and may operate in the first mode (e.g., switch to the first mode) when the motor current I.sub.MOTOR

required by the motor drive circuit to rotate the motor is below the current threshold I.sub.TH. An example of when this might occur is during the movement of the covering material between a fully-lowered position to a fully-raised position where the magnitude of the motor current I.sub.MOTOR needed to drive the motor might be above the current threshold I.sub.TH for at least an initial period of movement of the covering material, but when the covering material is close to the fully-raised position, the magnitude of the motor current I.sub.MOTOR may reduce below the current threshold I.sub.TH and the control circuit may operate in the first mode to drive the motor from the batteries (e.g., directly from the batteries). In some examples, the motor current I.sub.MOTOR may cross the current threshold I.sub.TH during some, but not all movements of the covering material. Further, in some examples, the motor current I.sub.MOTOR may cross the current threshold I.sub.TH towards the end of the movement of the covering material, and/or multiple times during the movement of the covering material. For instance, some movements of the covering material may have a profile for the magnitude of the motor current I.sub.MOTOR that looks like a parabola (or upside down parabola), for example, in instances where the motorized window treatments includes a torsion spring. In such examples, the magnitude of the motor current I.sub.MOTOR may cross the current threshold I.sub.TH multiple times during a single movement of the covering material.
(136) Although described primarily in the contact of a motorized window treatment that includes a motor for moving a covering material to control an amount of daylight entering a space, the methods, systems, and apparatuses described herein may be used with any load types, particularly loads that draw high peaks of current for relatively short periods of time and relatively infrequently throughout the day (e.g., motors, exhaust fans, elevators, lifts, emergency lighting, lights on for a short period of time, such as egress lightings, microwaves or other small appliances, etc.).

Claims

1. A motor drive unit for a motorized window treatment, the motor drive unit comprising: a bus capacitor configured to store a bus voltage; a motor configured to control movement of a covering material of the motorized window treatment; a motor drive circuit configured to receive the bus voltage and conduct a motor current through the motor for controlling power delivered to the motor to control movement of the covering material; a first switching circuit configured to be coupled between the bus capacitor and a first power source that is configured to generate a first power source voltage; a second power source configured to generate a second power source voltage; a second switching circuit coupled between the bus capacitor and the second power source; and a control circuit configured to control the first and second switching circuits, wherein, prior to controlling the motor drive circuit to conduct the motor current through the motor to control the movement of the covering material, the control circuit is configured to: when a magnitude of the second power source voltage is greater than a movement capacity threshold, render conductive the second switching circuit to charge the magnitude of the bus voltage to the magnitude of the second power source voltage, and control the motor drive circuit to conduct the motor current from the second power source and through the motor to control the movement of the covering material; when the magnitude of the second power source voltage is less than the movement capacity threshold, gradually render conductive the first switching circuit to charge the magnitude of the bus voltage to the magnitude of the first power source voltage, and control the motor drive circuit to conduct the motor current from the first power source and through the motor to control the movement of the covering material; and when movement of the covering material is complete, render non-conductive at least one of the first switching circuit or the second switching circuit that was rendered conductive to control the motor.
2. The motor drive unit of claim 1, wherein, to gradually render conductive the first switching circuit, the control circuit is configured to generate a pulse width modulated (PWM) gate signal at a

gate of the first switching circuit.
3. The motor drive unit of claim 2, wherein the control circuit is configured to increase an on-time of the PWM gate signal from one period to the next while gradually rendering conductive the first switching circuit.
4. The motor drive unit of claim 2, wherein the control circuit is configured to generate the PWM gate signal to render conductive the first switching circuit using open-loop control.
5. The motor drive unit of claim 1, wherein, to gradually render conductive the first switching circuit, the control circuit is configured to pulse width modulate a first switch control signal, wherein the first switch control signal is configured to render the first switching circuit conductive and non-conductive.
6. The motor drive unit of claim 1, wherein, to gradually render conductive the first switching circuit, the control circuit is configured to decrease an impedance of the first switching circuit from a non-conductive impedance to a conductive impedance.
7. The motor drive unit of claim 6, wherein the non-conductive impedance of the first switching circuit is greater than the conductive impedance of the first switching circuit.
8. The motor drive unit of claim 1, wherein the first switching circuit comprises at least one field-effect transistor (FET); and wherein, to gradually render conductive the first switching circuit, the control circuit is configured to control an impedance of the FET of the first switching circuit in a linear region.
9. The motor drive unit of claim 1, further comprising: a filter circuit coupled in series between the first switching circuit and the bus capacitor, the filter circuit configured to filter the motor current conducted through the first power source when the first switching circuit is conductive and the motor drive circuit is controlling the power delivered to the motor.
10. The motor drive unit of claim 9, wherein the filter circuit comprises an inductor.
11. The motor drive unit of claim 10, further comprising: a diode coupled between circuit common and a junction of the first switching circuit and the filter circuit, the diode configured to conduct current through the inductor and the bus capacitor when the first switching circuit is non-conductive while the switching circuit is gradually rendered conductive.
12. The motor drive unit of claim 9, wherein the filter circuit is configured to filter the motor current to conduct a first power source current through the first power source that has a DC magnitude.
13. The motor drive unit of claim 1, wherein the motor drive unit is configured such that the first switching circuit and the second switching circuit cannot both be rendered conductive at the same time.
14. The motor drive unit of claim 1, wherein the first power source comprises one or more alkaline batteries, and the second power source comprises one or more lithium batteries or supercapacitors.
15. The motor drive unit of claim 1, wherein the first power source comprises one or more replaceable batteries, and the second power source comprises one or more rechargeable batteries or supercapacitors.
16. The motor drive unit of claim 1, wherein the first power source is characterized by a larger equivalent series resistance than the second power source.
17. The motor drive unit of claim 1, wherein the first power source comprises a solar energy receiving circuit, an ultrasonic energy receiving circuit, or a radio-frequency (RF) energy receiving circuit.
18. The motor drive unit of claim 1, wherein the first power source is removably replaceable by a user.
19. The motor drive unit of claim 1, wherein the movement capacity threshold indicates a storage level sufficient to complete a full movement of the covering material from a fully-lowered position to a fully-raised position.
20. The motor drive unit of claim 1, wherein the first switching circuit comprises at least one field-

effect transistor (FET).
21. The motor drive unit of claim 1, further comprising: a power converter circuit configured to charge the second power source from the first power source voltage to produce the second power source voltage across the second power source; and wherein the control circuit is configured to: set a charging flag in response to enabling the power converter circuit, and clear the charging flag in response to disabling the power converter circuit.
22. The motor drive unit of claim 21, wherein the control circuit is configured to: enable the power converter circuit when the second power source voltage is less than a low-side threshold; and disable the power converter circuit when the second power source voltage is greater than a high-side threshold, wherein the movement capacity threshold is less than the low-side threshold and the high-side threshold.
23. The motor drive unit of claim 21, wherein the control circuit is configured to: wait a delay period after controlling the motor when movement of the covering material is complete before enabling the power converter circuit to charge the second power source from the first power source voltage.
24. The motor drive unit of claim 21, wherein the control circuit is configured to use a different delay period based on whether the first switching circuit or the second switching circuit was render conductive during the movement of the covering material.
25. The motor drive unit of claim 1, wherein the control circuit is configured to: set a lockout flag in response to a reception of a command to move the covering material of the motorized window treatment and a determination that the magnitude of the second power source voltage is less than the movement capacity threshold; and clear the lockout flag in response to a determination that the magnitude of the second power source voltage is greater than a high-side threshold.
26. The motor drive unit of claim 25, wherein the control circuit is configured to: gradually render conductive the first switching circuit to charge the magnitude of the bus voltage to approximately the magnitude of the first power source voltage when the lockout flag is set.
27. The motor drive unit of claim 25, wherein the control circuit is configured to: gradually render conductive the first switching circuit to charge the magnitude of the bus voltage to approximately the magnitude of the first power source voltage when the lockout flag is set regardless of the magnitude of the second power source voltage.